Signals and Systems

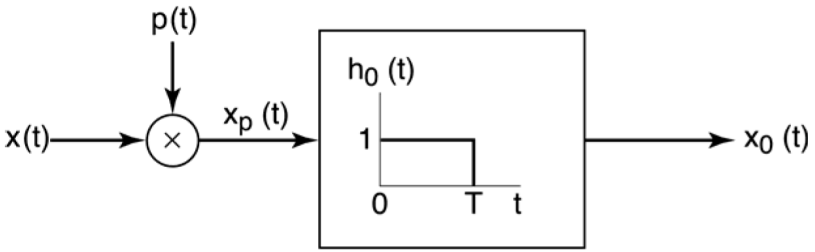
Lecture 17: Sampling and Quantization

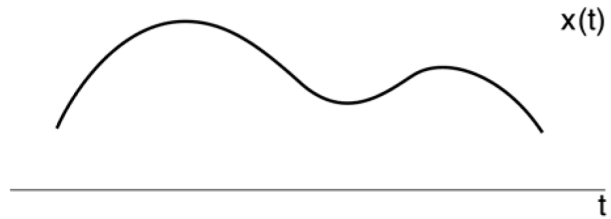
Instructor: Ting Zhang
Zhejiang University

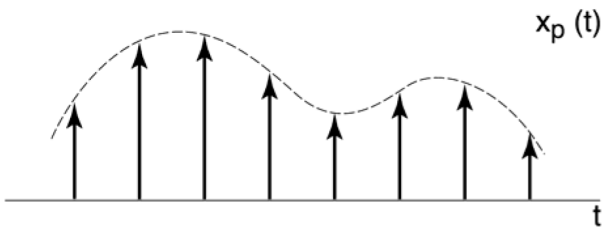
05/22/2025
Adapted from the materials provided on
the MIT OpenCourseWare

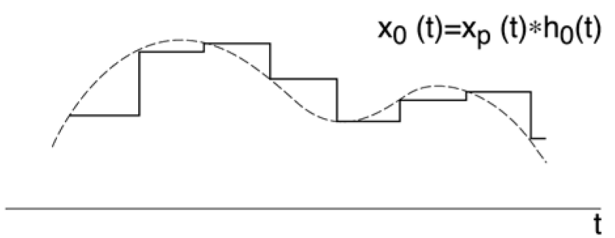
Observations on Sampling

(1) In practice, we obviously don't sample with impulses or implement ideal lowpass filters.
   — One practical example: The Zero-Order Hold



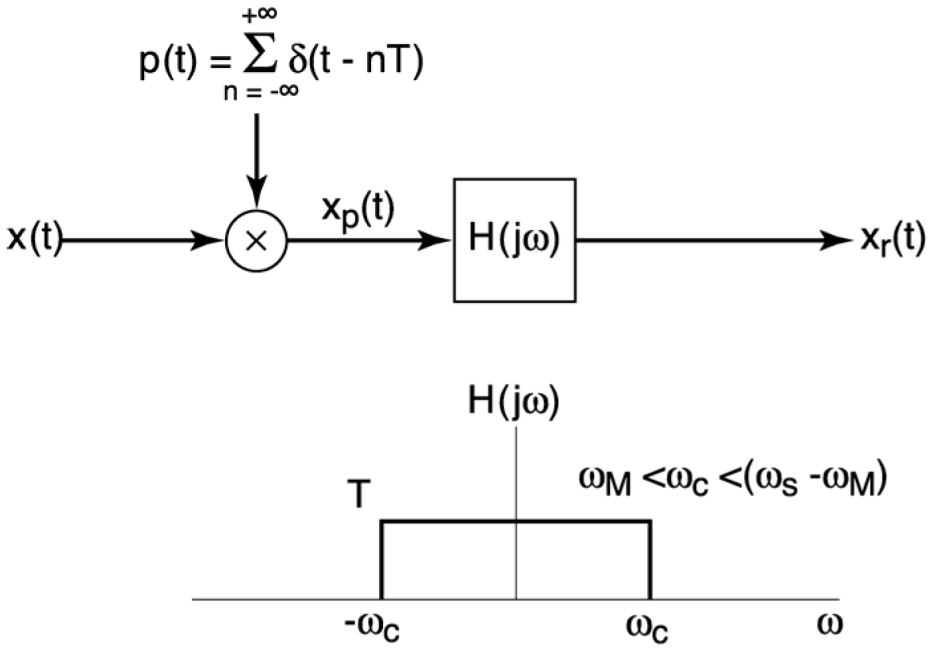






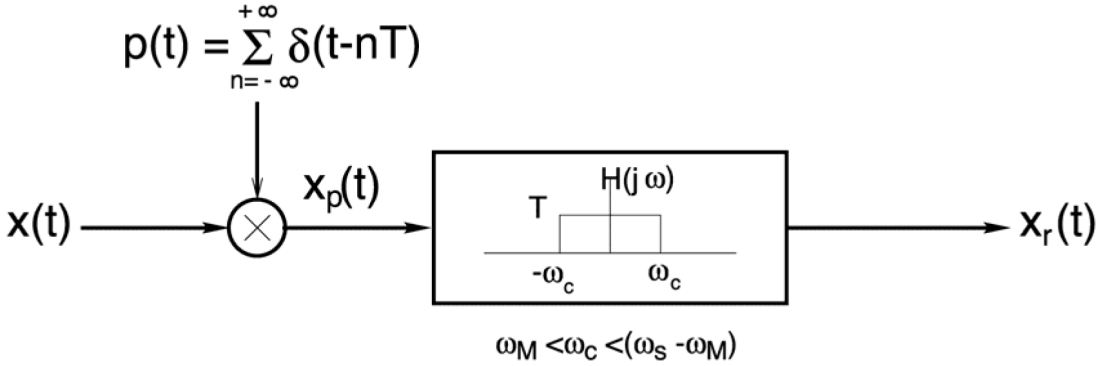
Observations (Continued)

(2) Sampling is fundamentally a *time-varying* operation, since we multiply $x(t)$ with a time-varying function $p(t)$. However,



is the identity system (which is *TI*) for bandlimited $x(t)$ satisfying the sampling theorem ($\omega_s > 2\omega_M$).

Time-Domain Interpretation of Reconstruction of Sampled Signals — Band-Limited Interpolation



$$x_r(t) = x_p(t) * h(t) \quad , \quad \text{where } h(t) = \frac{T \sin \omega_c t}{\pi t}$$

$$= \left(\sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT) \right) * h(t)$$

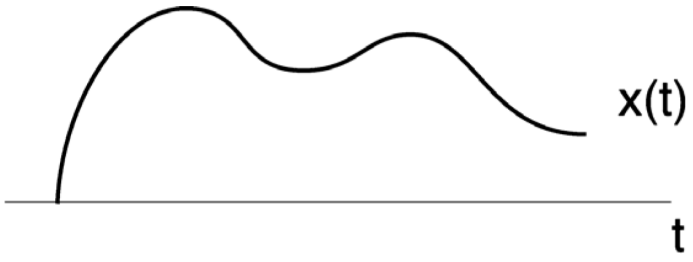
$$= \sum_{n=-\infty}^{\infty} x(nT)h(t - nT) = \sum_{n=-\infty}^{\infty} x(nT) \frac{T \sin[\omega_c(t - nT)]}{\pi(t - nT)}$$

The lowpass filter interpolates the samples *assuming* $x(t)$ contains no energy at frequencies $\geq \omega_c$

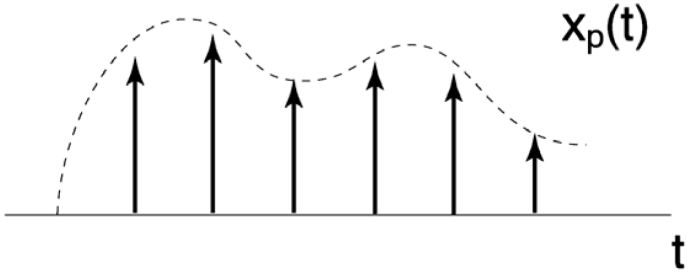
Interpolation

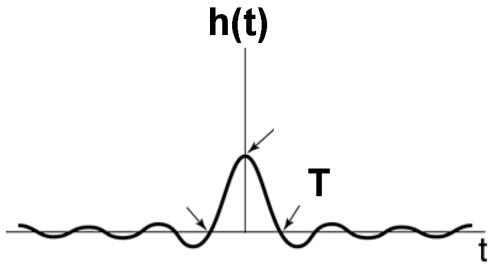
Graphic Illustration of Time-Domain Interpolation

Original
CT signal

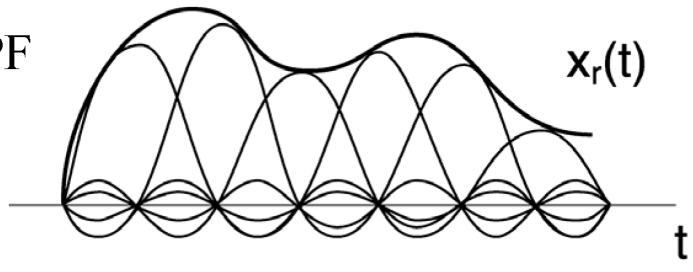


After sampling



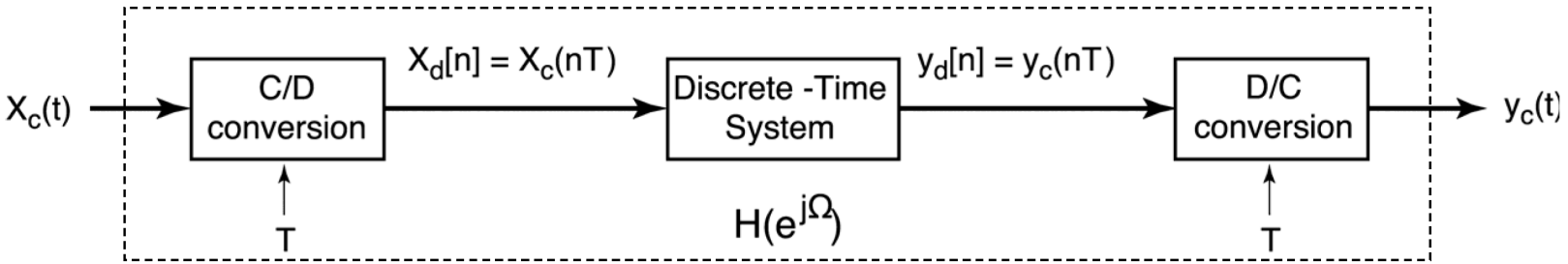


After passing the LPF



Discrete-Time Processing of CT Signals

DT Processing of *Band-Limited* CT Signals



Why do this?

— Inexpensive, versatile, and higher noise margin.

How do we analyze this system?

— We will need to do it in the frequency domain in both CT *and* DT

— In order to avoid confusion about notations, specify

ω — CT frequency variable

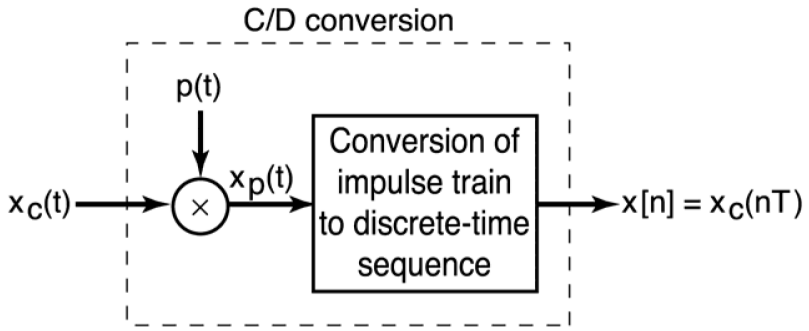
Ω — DT frequency variable ($\Omega = \omega T$)

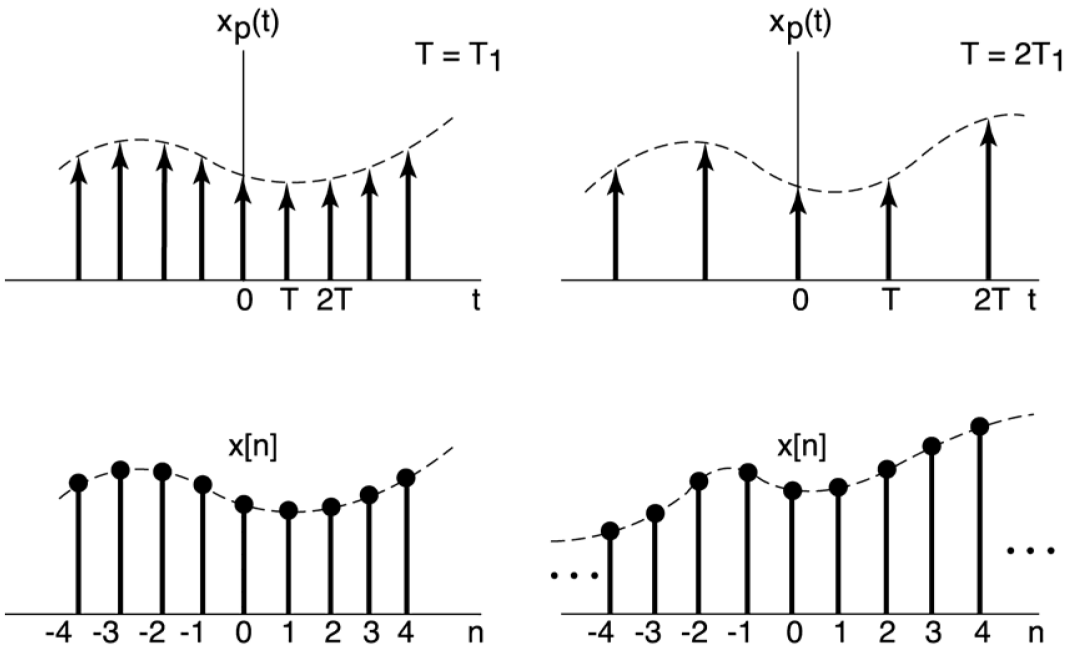
Step 1: Find the relation between $x_c(t)$ and $x_d[n]$, or $X_c(j\omega)$ and $X_d(e^{j\Omega})$

Discrete-Time Processing of CT Signals

Time-Domain Interpretation of C/D Conversion

Note: Not full analog/digital (A/D) conversion – not quantizing the x[n] values





Frequency-Domain Interpretation of C/D Conversion

$$x_p(t) = x_c(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} x_c(nT)\delta(t - nT)$$

$$\Downarrow \mathcal{F}$$

$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\omega - k\omega_s)) = \sum_{n=-\infty}^{\infty} x_c(nT)e^{-j\omega nT} \quad (1)$$

CT – (periodic with period $\omega_s = 2\pi/T$)

$$X_d(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x_d[n]e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} x_c(nT)e^{-j\Omega n} \quad (2)$$

DT – (periodic with period 2π)

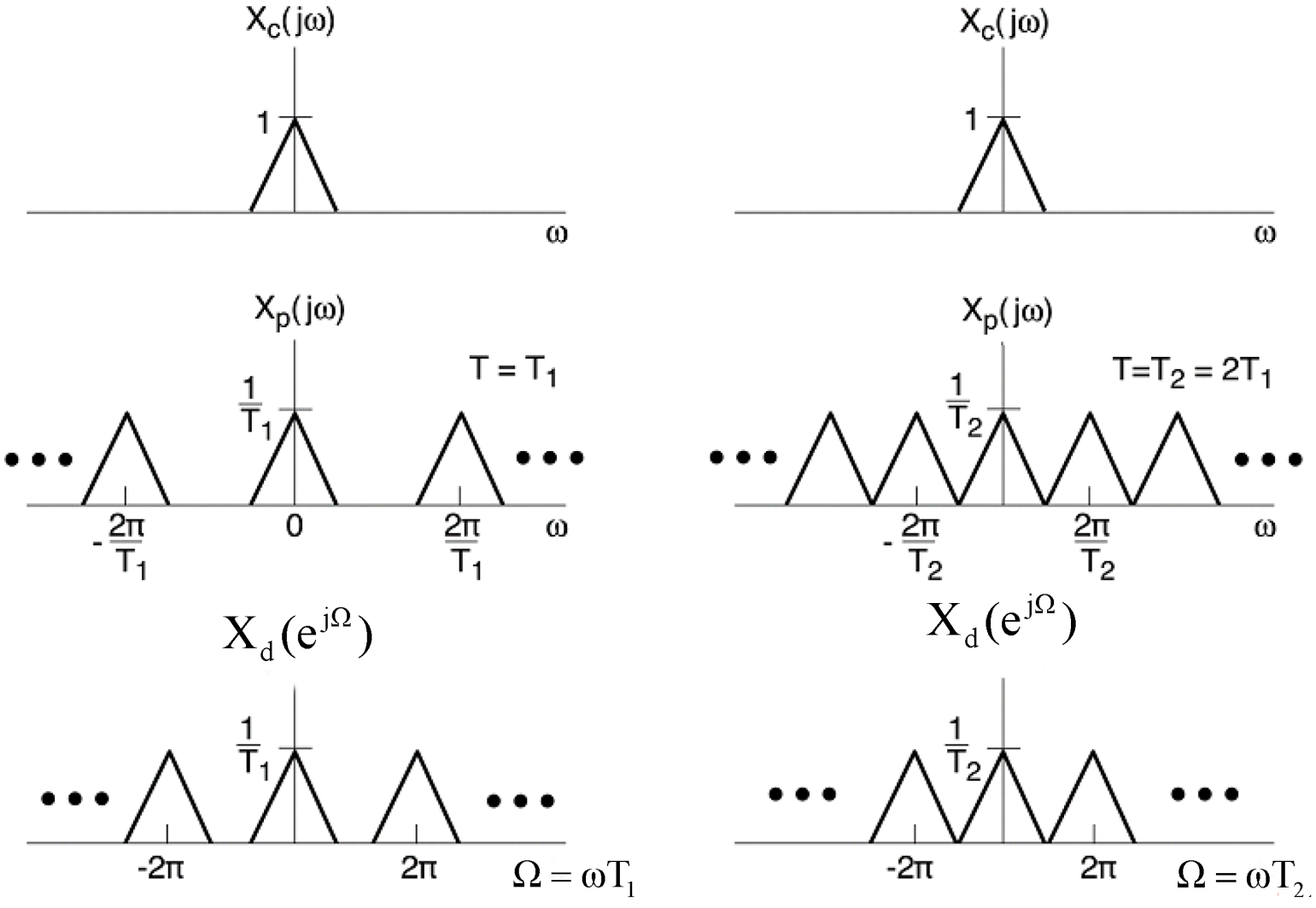
$\Downarrow$ Compare Eqs. (1) & (2) and note $\Omega = \omega T$

$$X_d(e^{j\Omega}) = X_p\left(j\left(\frac{\Omega}{T}\right)\right)$$

Note: $\omega_s \Leftrightarrow 2\pi$
CT DT

Discrete-Time Processing of CT Signals

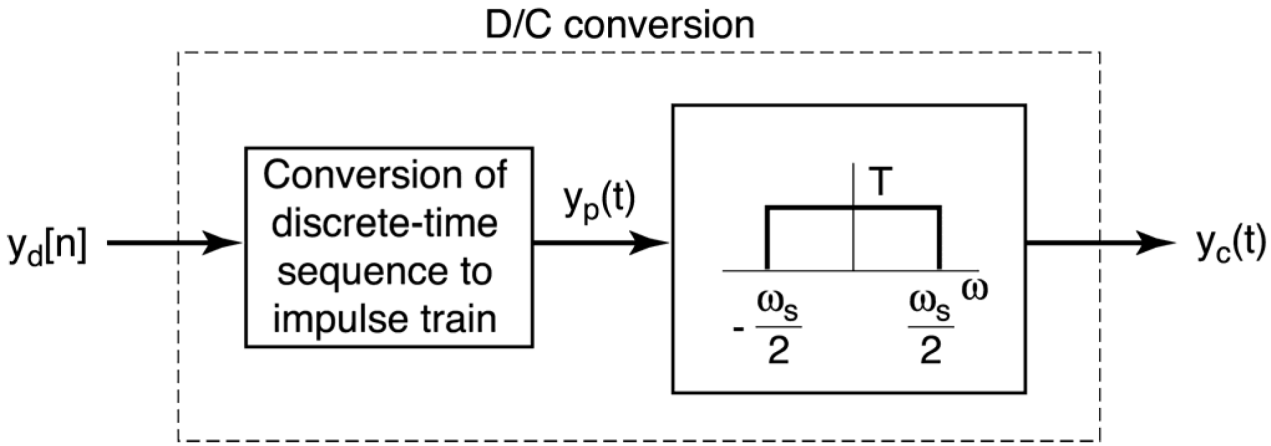
Illustration of C/D Conversion in the Frequency-Domain



Discrete-Time Processing of CT Signals

D/C Conversion $y_d[n] \rightarrow y_c(t)$

Reverse of the process of C/D conversion



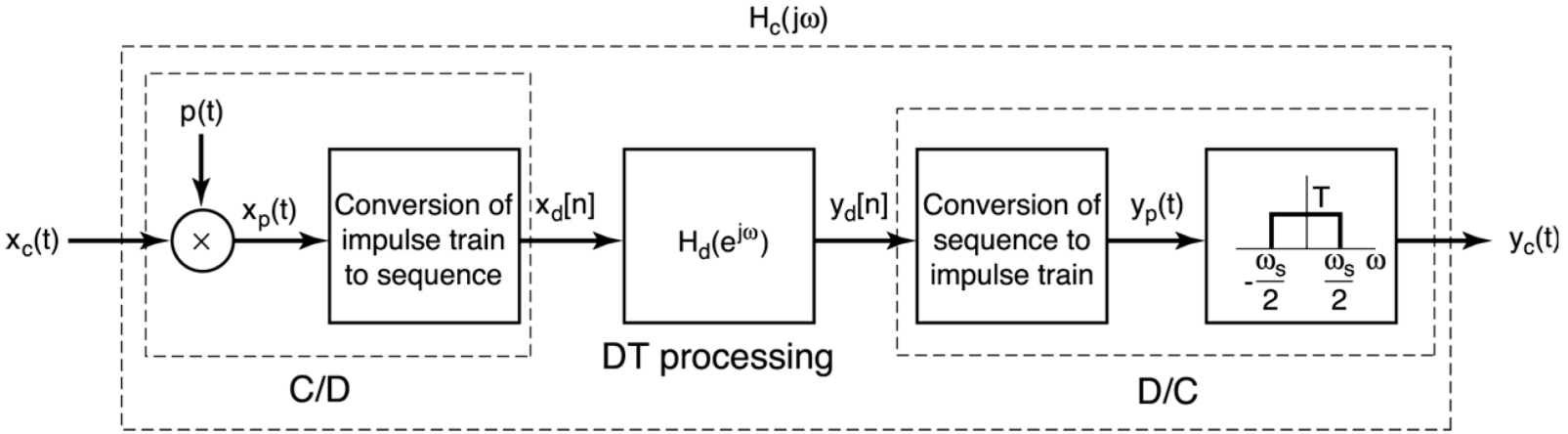
Again, $\Omega = \omega T$

$Y_p(j\omega) = Y_d(e^{j\omega T})$ — Reverses frequency scaling

$$Y_c(j\omega) = \begin{cases} TY_d(e^{j\omega T}), & |\omega| < \frac{\omega_s}{2} - \text{ bandlimited} \\ 0, & \text{otherwise} \end{cases}$$

Discrete-Time Processing of CT Signals

Now the whole picture

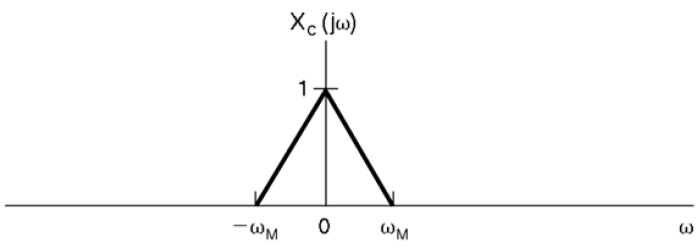


- Overall system is time-varying if sampling theorem is *not* satisfied
- It is LTI if the sampling theorem *is satisfied*, i.e. for bandlimited inputs $x_c(t)$, with $\omega_M < \frac{\omega_s}{2}$
- When the input $x_c(t)$ is band-limited ($X(j\omega) = 0$ at $|\omega| > \omega_M$) and the sampling theorem is satisfied ($\omega_s > 2\omega_M$), then

$$Y_c(j\omega) = H_c(j\omega)X_c(j\omega) \longleftrightarrow y_c(t) = h_c(t) * x_c(t) \quad \text{LTI}$$

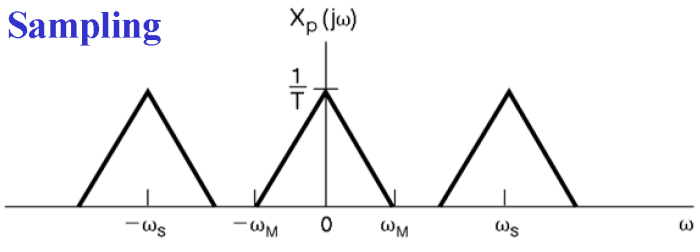
Discrete-Time Processing of CT Signals

Frequency-Domain Illustration of DT Processing of CT Signals



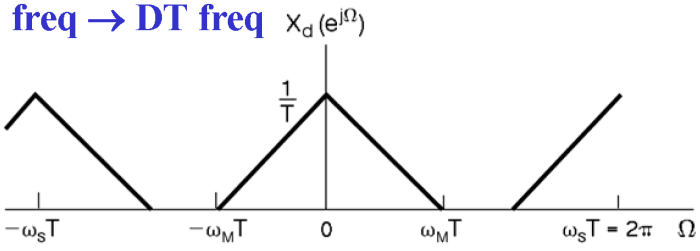
(a)

Sampling

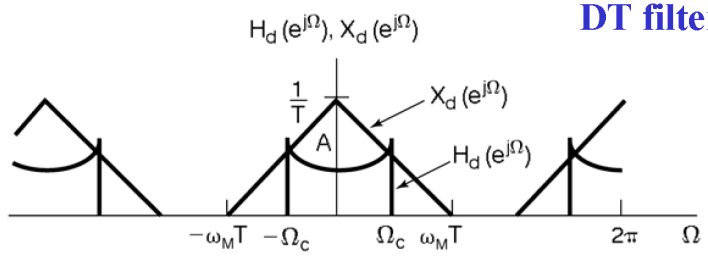


(b)

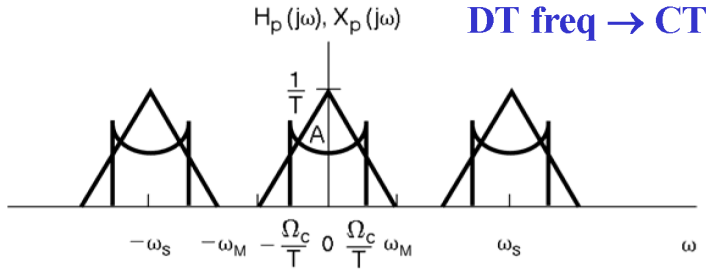
CT freq → DT freq



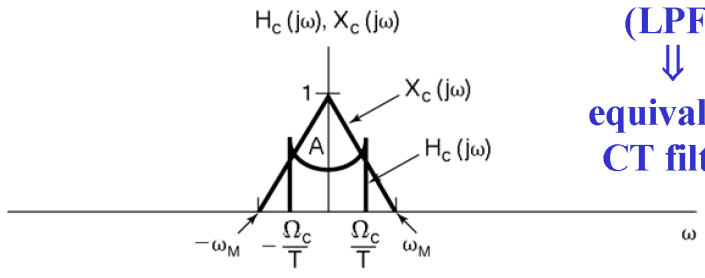
(c)



(d)



(e)



(f)

Interpolate (LPF)
⇓
equivalent CT filter

Discrete-Time Processing of CT Signals

Assuming No Aliasing

Step 1 – C/D: $X_d(e^{j\Omega}) = X_p(j\Omega/T)$ – periodic

$\omega = \Omega/T$ $= \frac{1}{T} X_c(j\Omega/T), \quad -\pi < \Omega < \pi$ if no aliasing

Step 2 – DT Filter: $Y_d(e^{j\Omega}) = H_d(e^{j\Omega}) X_d(e^{j\Omega})$

$= \frac{1}{T} H_d(e^{j\Omega}) X_c(j\Omega/T), \quad -\pi < \Omega < \pi$

Step 3 – D/C: $Y_c(j\omega) = \begin{cases} TY_d(e^{j\omega T}) = H_d(e^{j\omega T}) X_c(j\omega), & -\frac{\omega_s}{2} < \omega < \frac{\omega_s}{2} \\ 0, & \text{otherwise} \end{cases}$

$\Omega = \omega T$

$\Downarrow$

$$H_c(j\omega) = \begin{cases} H_d(e^{j\omega T}), & |\omega| < \frac{\omega_s}{2} \\ 0, & \text{otherwise} \end{cases}$$

In practice, first specify the desired $H_c(j\omega)$, then design $H_d(e^{j\Omega})$.

Check Yourself

Example: Digital Differentiator

Applications: Edge Enhancement

Courtesy of Jason Oppenheim.
Used with permission.

Courtesy of Jason Oppenheim.
Used with permission.

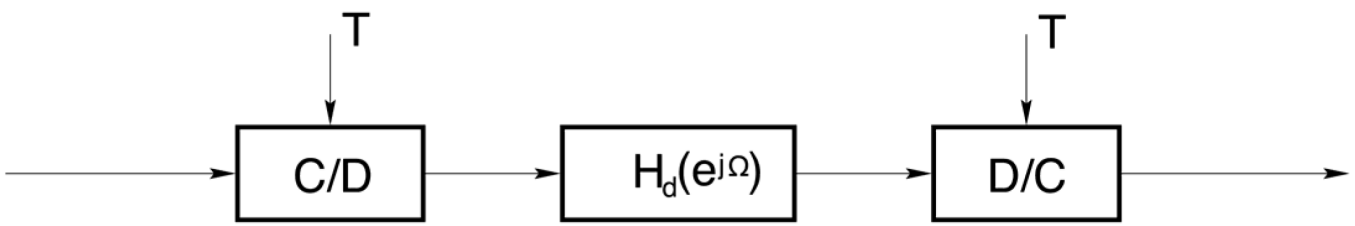
Check Yourself

Construction of Digital Differentiator

Bandlimited Differentiator

$$\text{Desired:} \quad H_c(j\omega) = \begin{cases} j\omega, & |\omega| < \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

Set $\omega_s = 2\omega_c \Rightarrow T = \frac{2\pi}{\omega_s} = \frac{\pi}{\omega_c}$. Assume $\omega_M < \omega_c$ (Nyquist rate met)



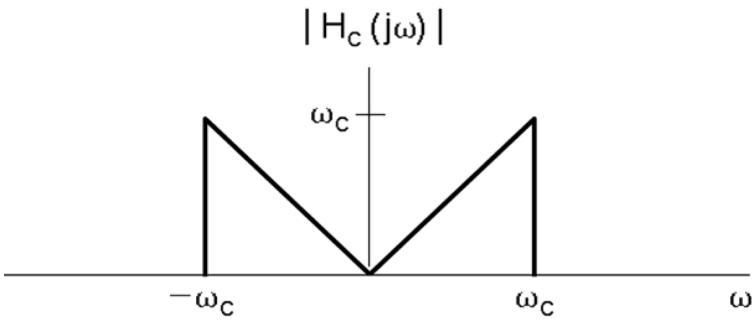
Choice for $H_d(e^{j\Omega})$:

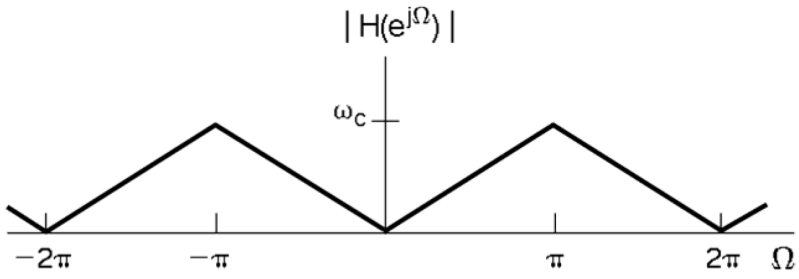
$$H_d(e^{j\Omega}) = \begin{cases} H_c(j\Omega/T), & |\Omega| < \pi \\ \text{periodic}, & |\Omega| \geq \pi \end{cases}$$

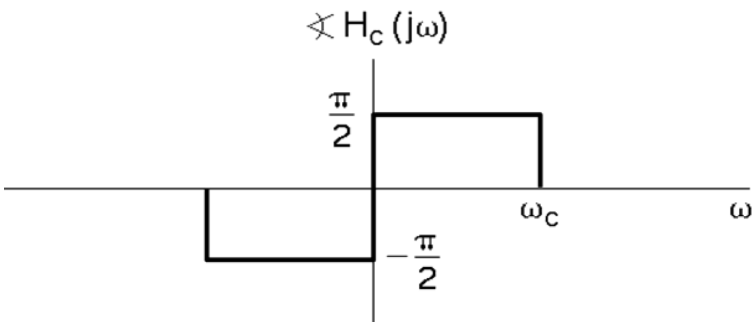
$$= j\left(\frac{\Omega}{T}\right) = j\omega_c\left(\frac{\Omega}{\pi}\right) \quad |\Omega| < \pi$$

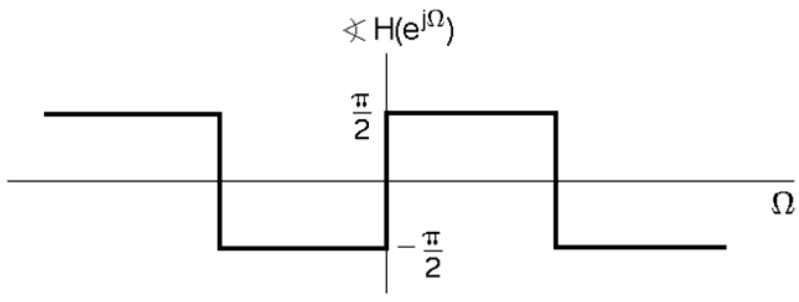
Check Yourself

Band-Limited Digital Differentiator (continued)







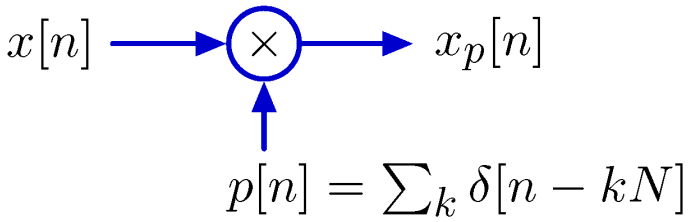


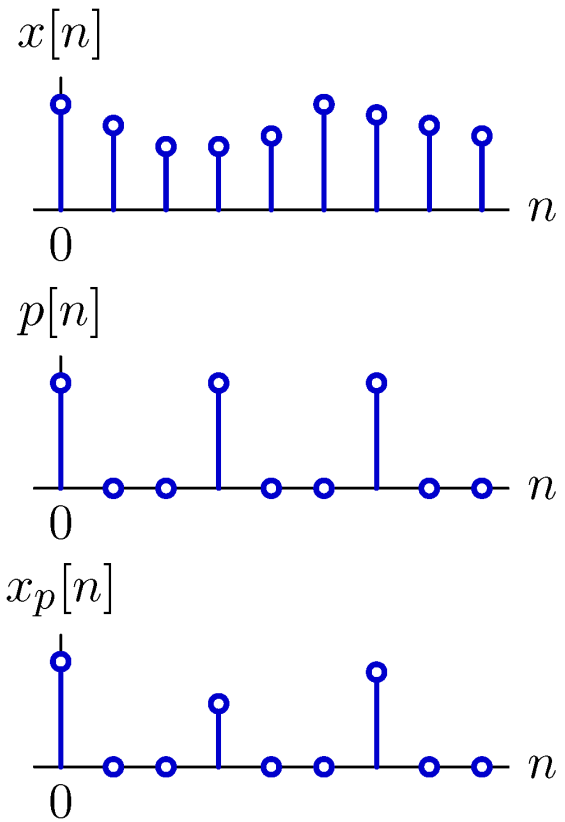
CT

DT

Discrete-Time Sampling (Resampling)

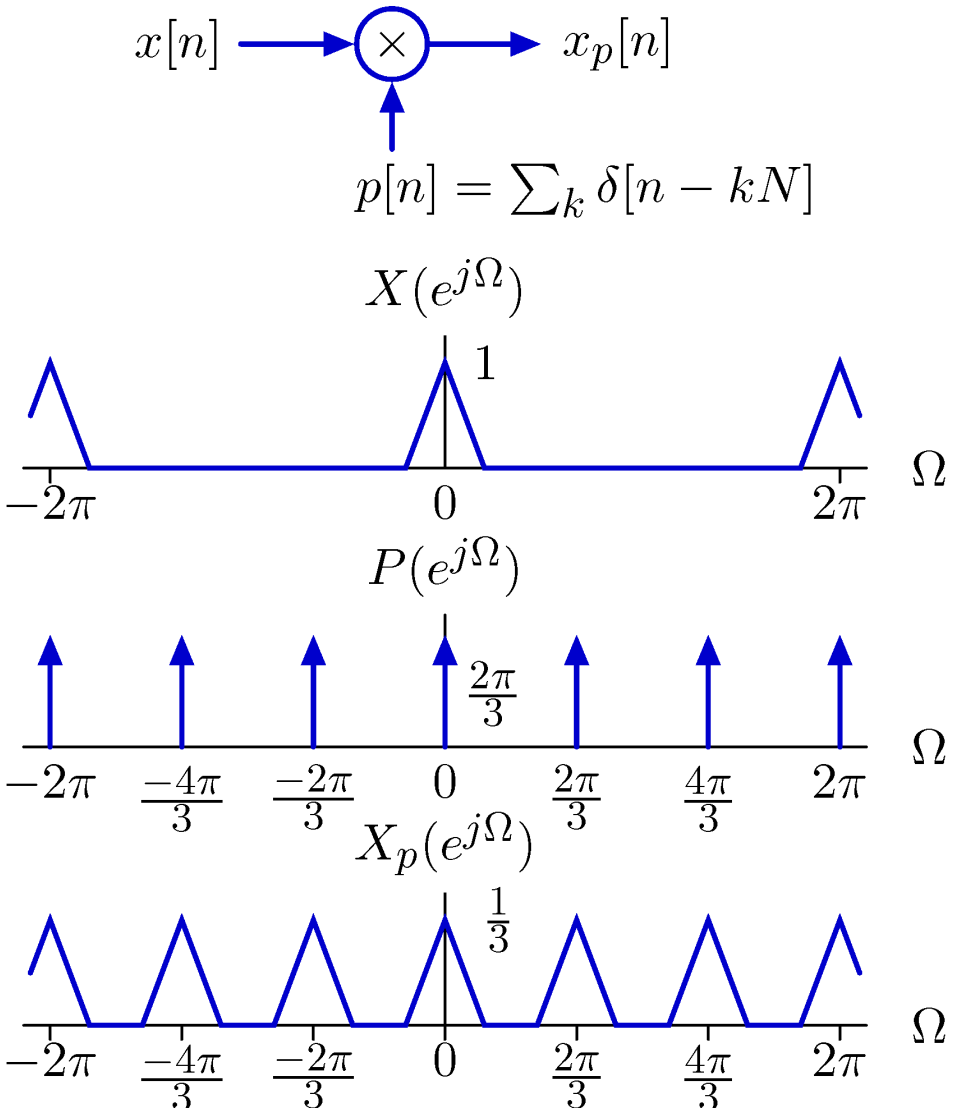
DT sampling is much like CT sampling.





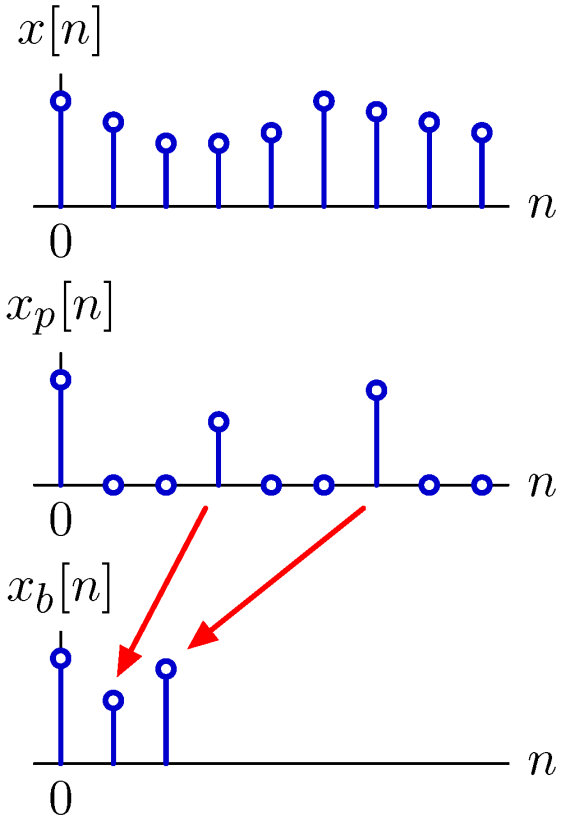
Discrete-Time Sampling

As in CT, sampling introduces additional copies of $X(e^{j\Omega})$.



Discrete-Time Sampling

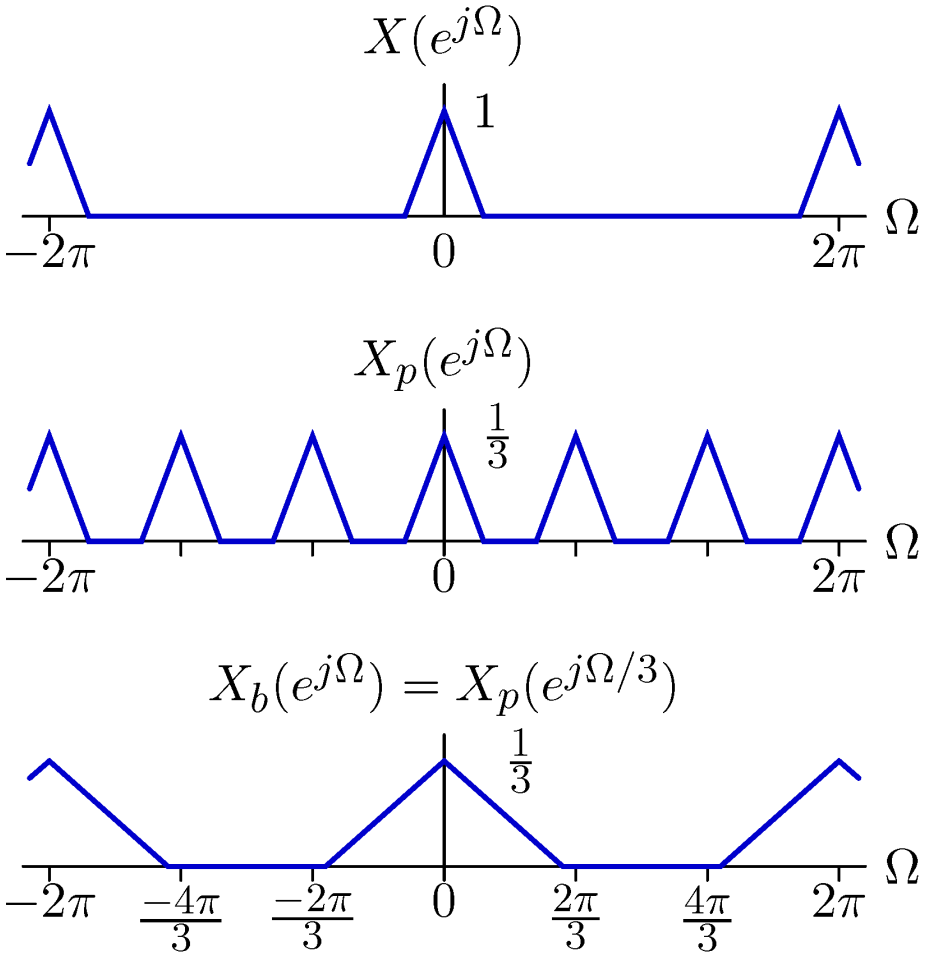
Sampling a finite sequence gives rise to a shorter sequence.



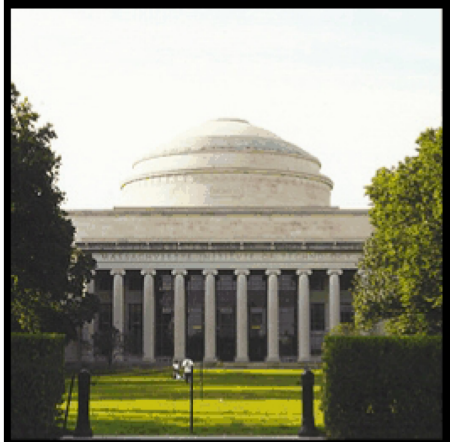
$$X_b(e^{j\Omega}) = \sum_n x_b[n]e^{-j\Omega n} = \sum_n x_p[3n]e^{-j\Omega n} = \sum_k x_p[k]e^{-j\Omega k/3} = X_p(e^{j\Omega/3})$$

Discrete-Time Sampling

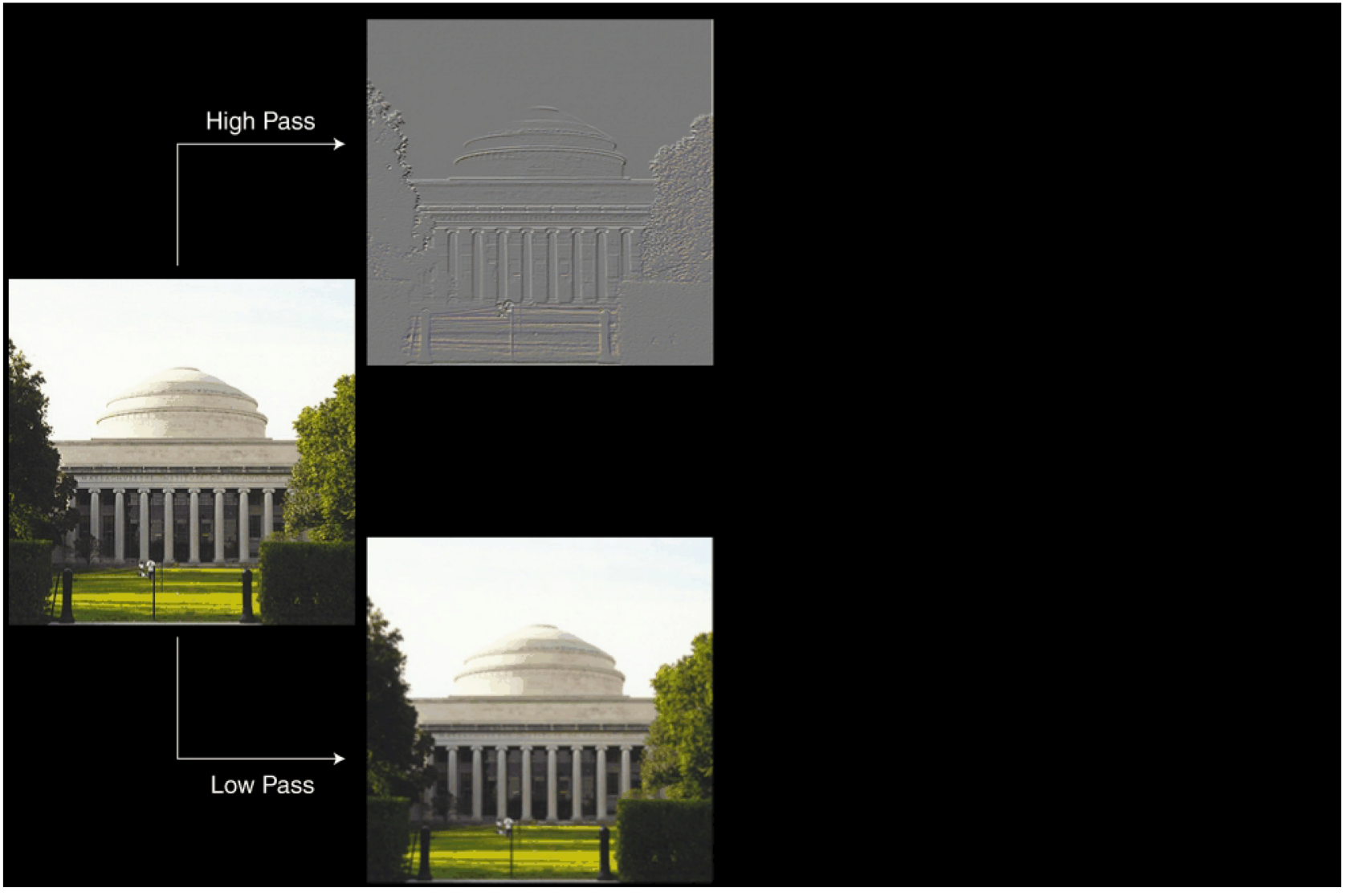
But the shorter sequence has a wider frequency representation.



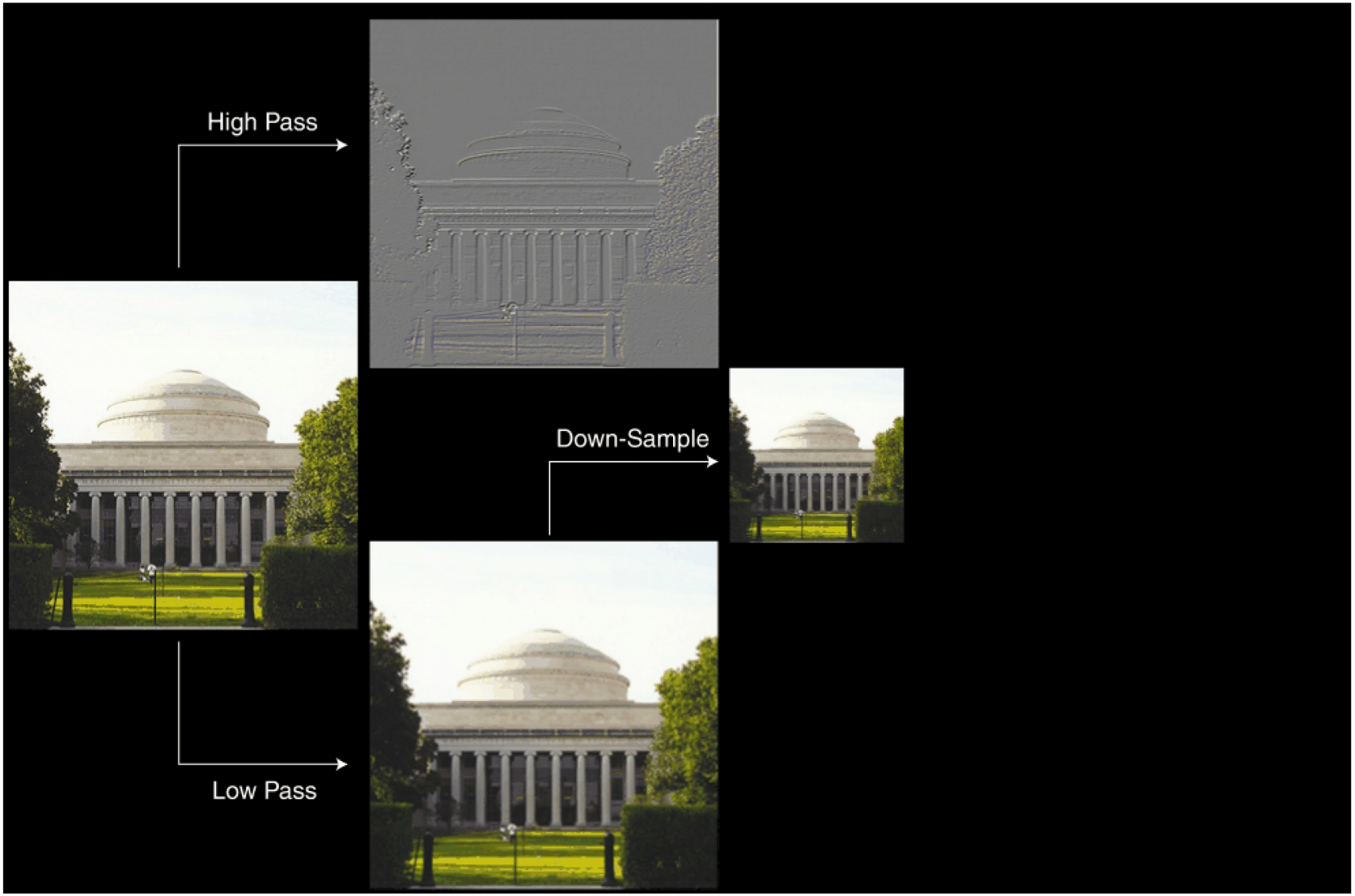
Discrete-Time Sampling



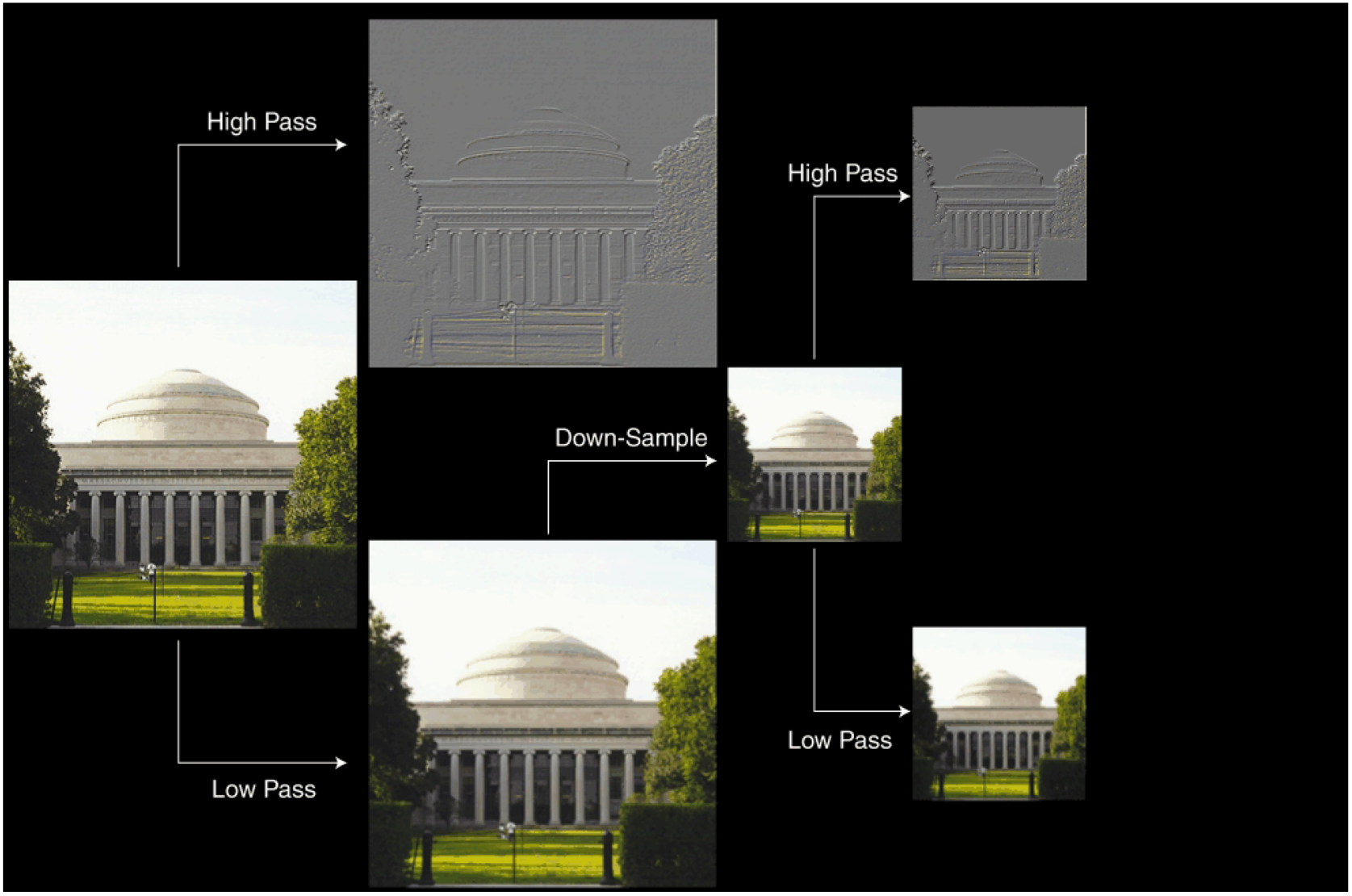
Discrete-Time Sampling



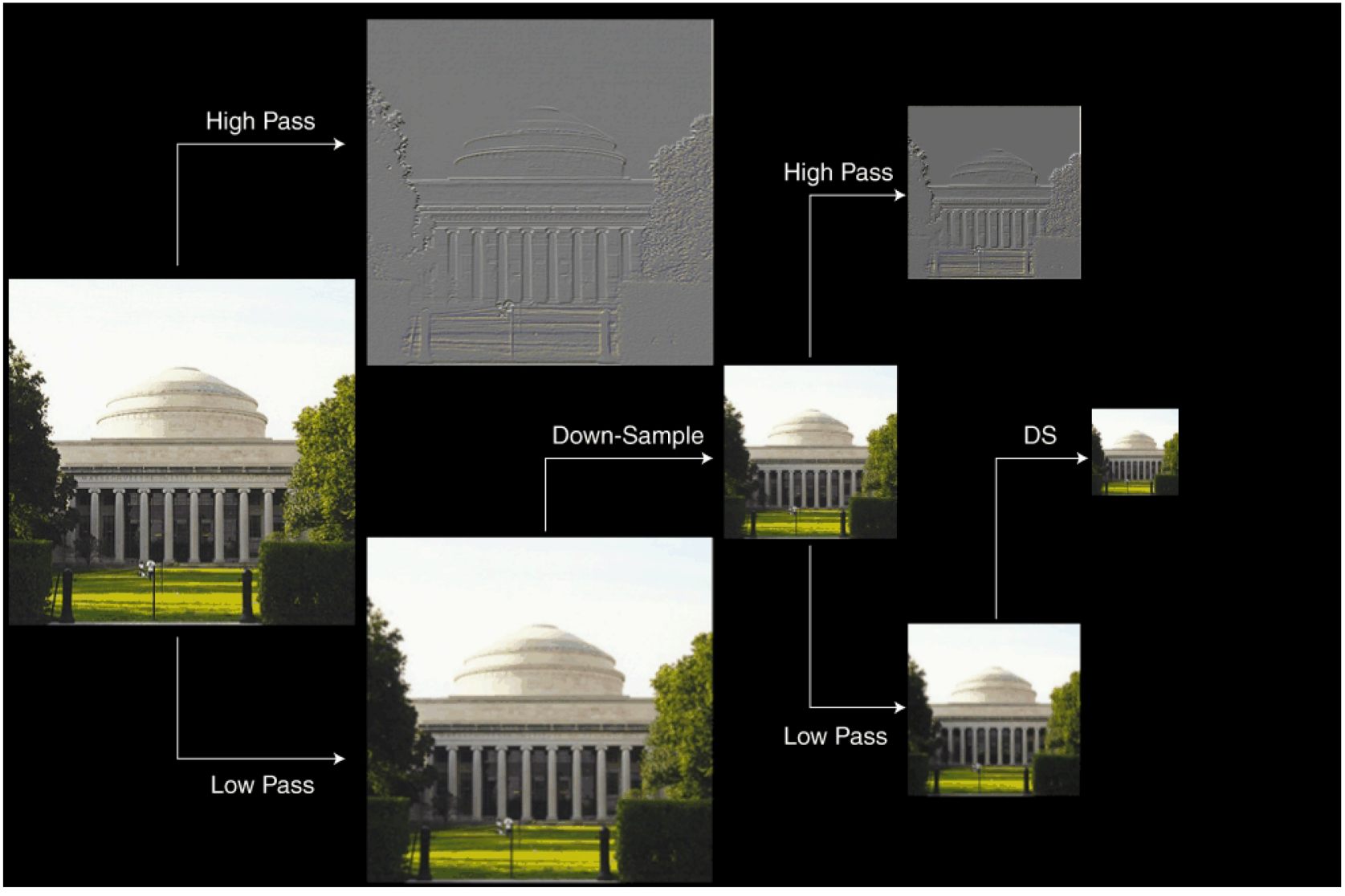
Discrete-Time Sampling



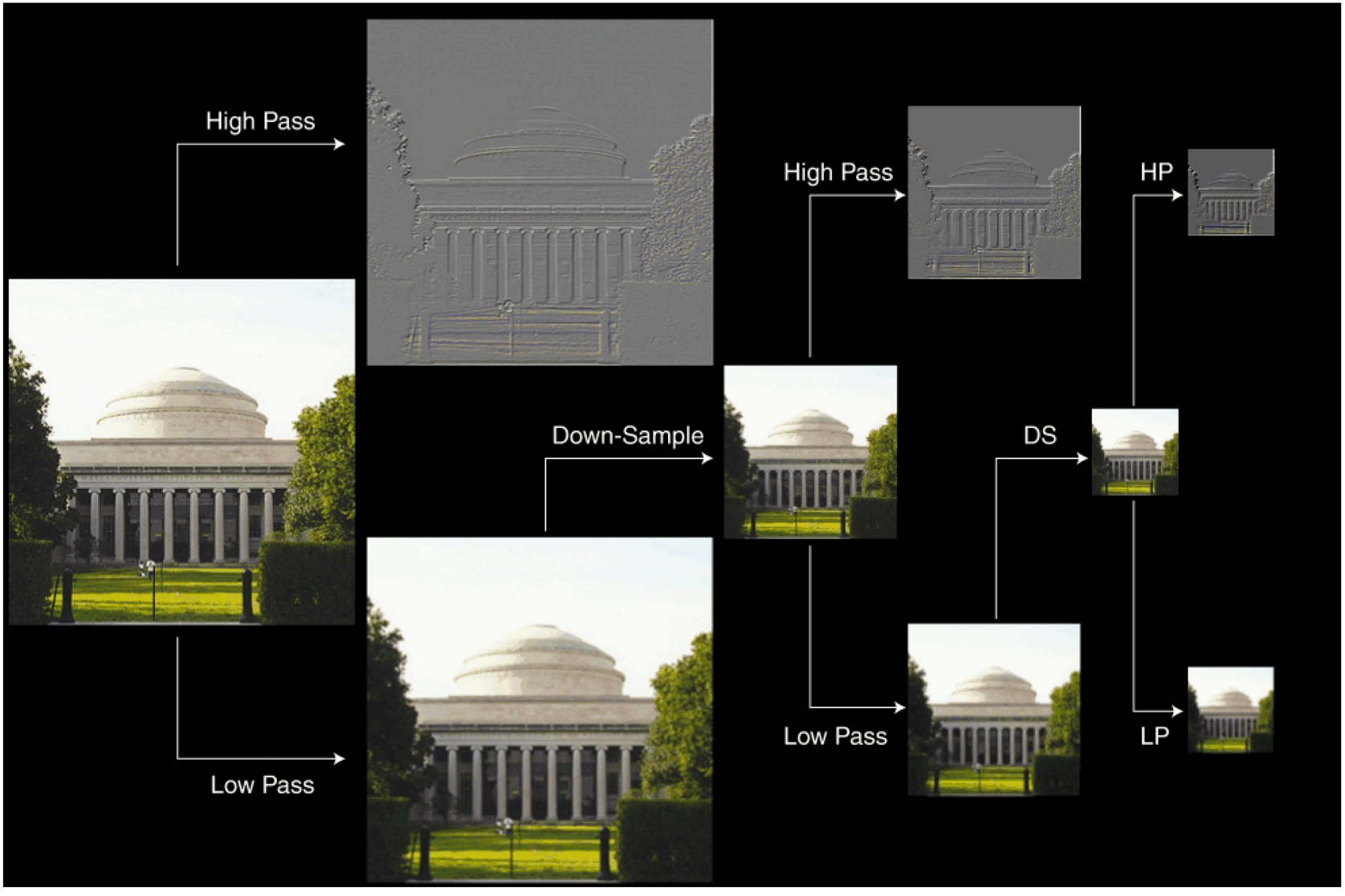
Discrete-Time Sampling



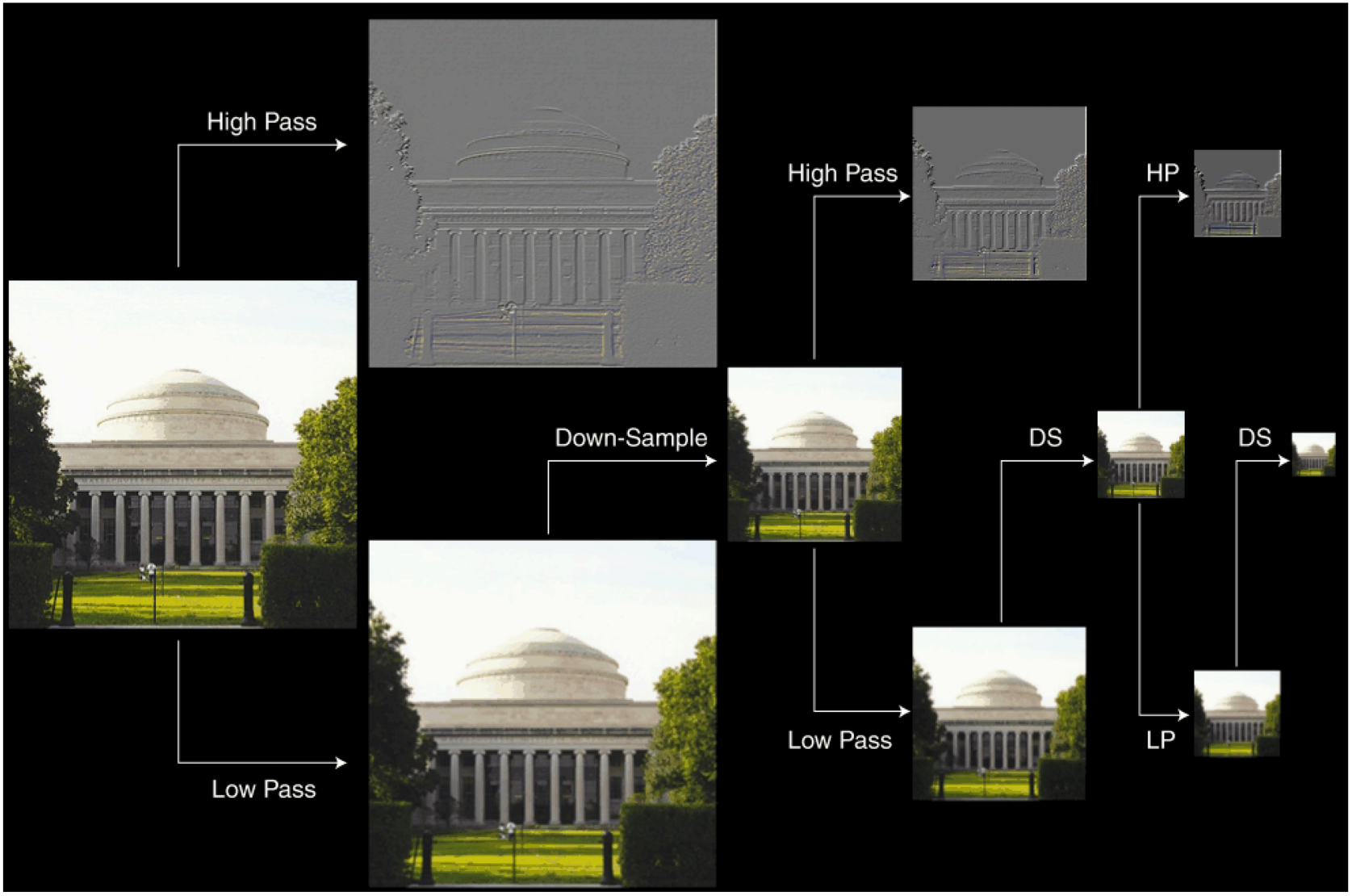
Discrete-Time Sampling



Discrete-Time Sampling



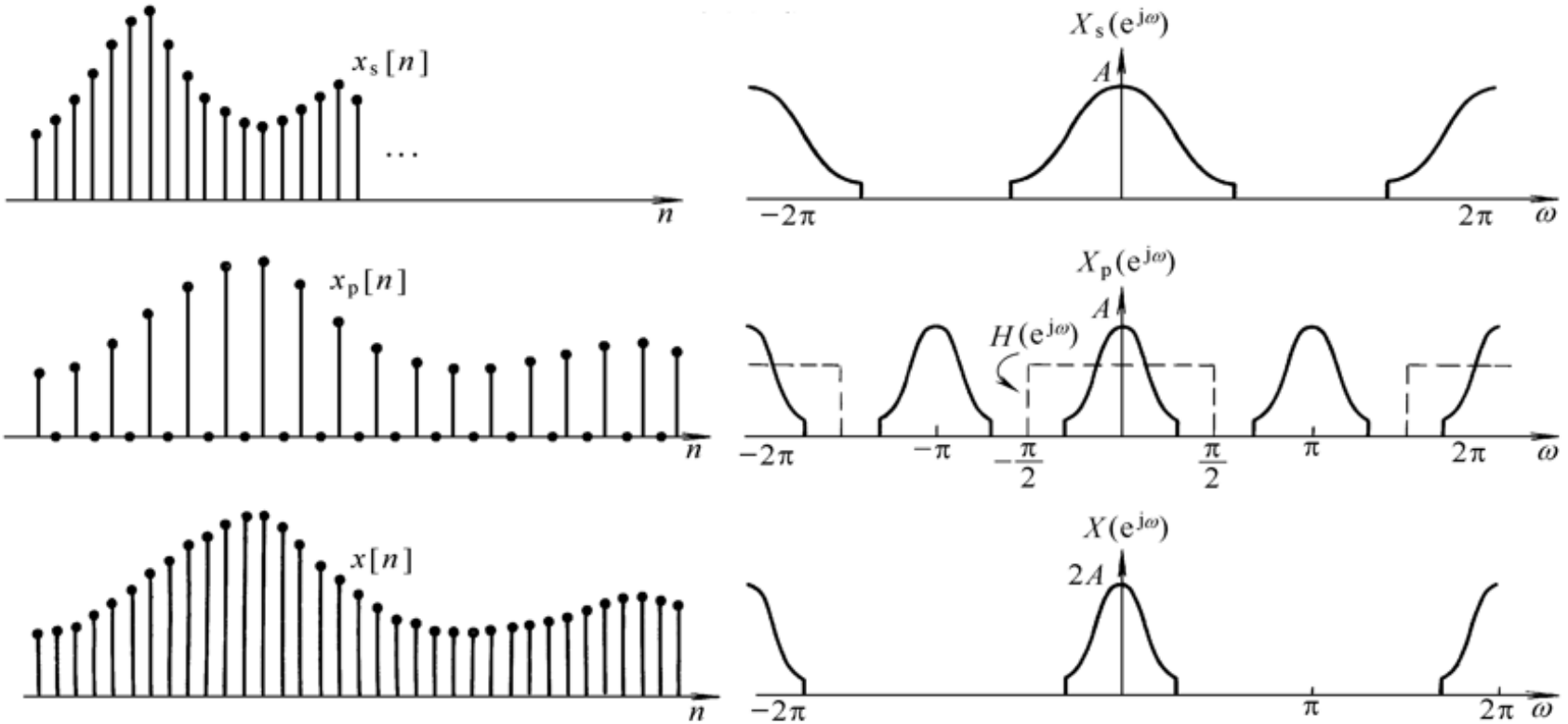
Discrete-Time Sampling



Discrete-Time Sampling: Upsampling

Insert zeros between samples to upsample the signal;

Then filter out the additional copies in frequency.



Discrete-Time Sampling: Upsampling

Discrete-Time Sampling: Upsampling

Discrete-Time Sampling: Upsampling

Discrete-Time Sampling: Upsampling

Discrete-Time Sampling: Upsampling

Quantization

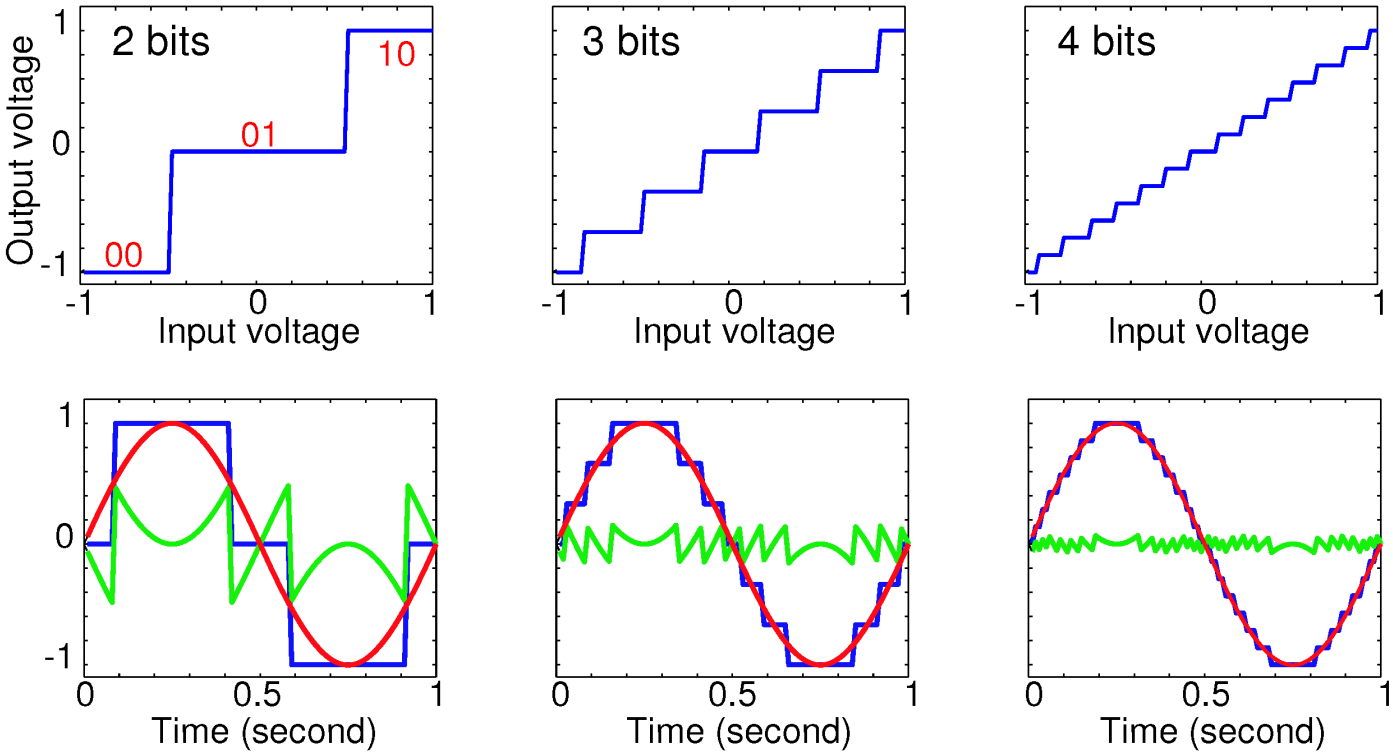
Digital recording, transmission, storage, and retrieval requires discrete representations of both time (e.g., sampling) and amplitude.

- audio: MP3, CD, cell phone
- pictures: digital camera, printer
- video: DVD
- everything on the web

Quantization: discrete representations for amplitudes

Quantization

We measure discrete amplitudes in bits.



Bit rate = (# bits/sample)×(# samples/sec)

Check Yourself

We hear sounds that range in amplitude from 1,000,000 to 1.

How many bits are needed to represent this range?

1. 5 bits
2. 10 bits
3. 20 bits
4. 30 bits
5. 40 bits

Check Yourself

How many bits are needed to represent 1,000,000:1?

bits	range
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1,024
11	2,048
12	4,096
13	8,192
14	16,384
15	32,768
16	65,536
17	131,072
18	262,144
19	524,288
20	1,048,576

Check Yourself

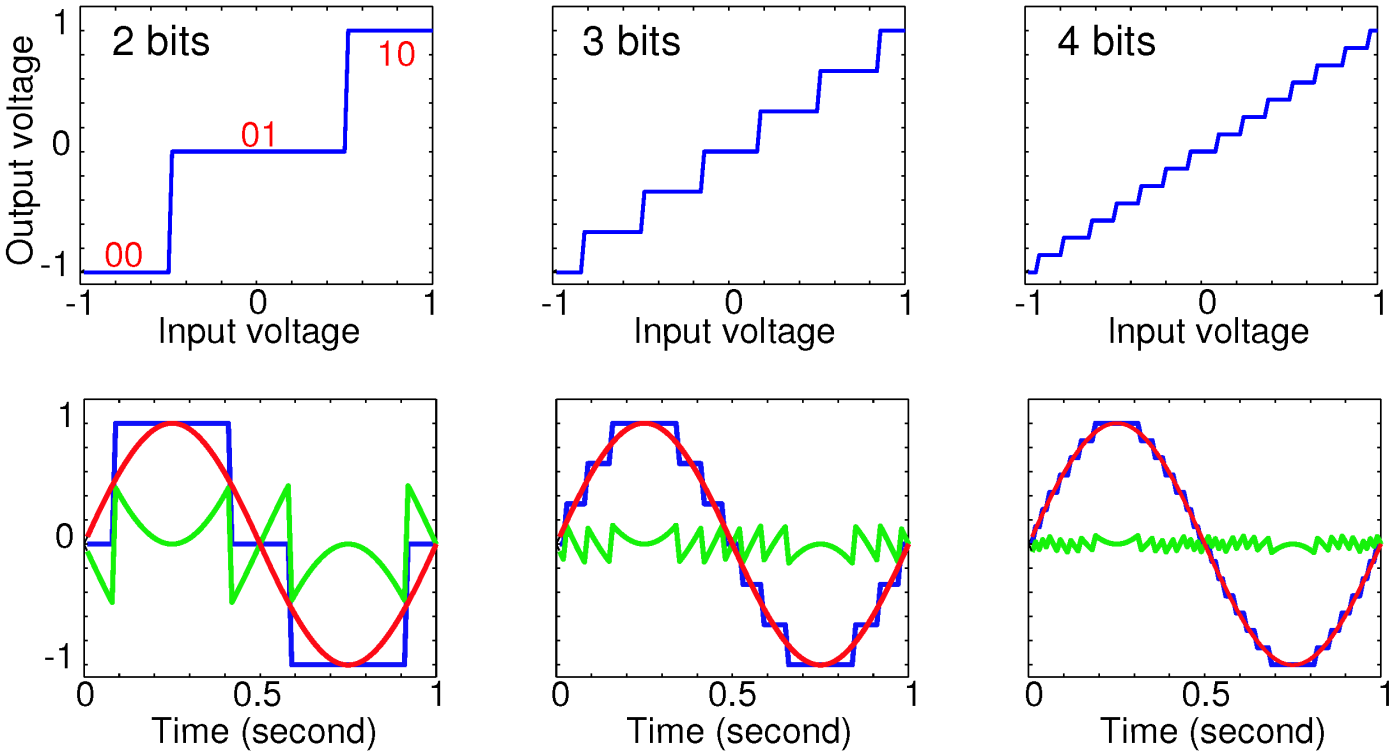
We hear sounds that range in amplitude from 1,000,000 to 1.

How many bits are needed to represent this range? 3

1. 5 bits
2. 10 bits
3. 20 bits
4. 30 bits
5. 40 bits

Quantization

We measure discrete amplitudes in bits.



Example: audio CD

$$2\,\text{channels} \times 16\,\frac{\text{bits}}{\text{sample}} \times 44,100\,\frac{\text{samples}}{\text{sec}} \times 60\,\frac{\text{sec}}{\text{min}} \times 74\,\text{min} \approx 6.3\,\text{G bits}$$

$$\approx 0.78\,\text{G bytes}$$

Quantizing Images

Converting an image from a continuous representation to a discrete representation involves the same sort of issues.

This image has 280×280 pixels, with brightness quantized to 8 bits.

Quantizing Images

8 bit image

7 bit image

Quantizing Images

8 bit image

6 bit image

Quantizing Images

8 bit image

5 bit image

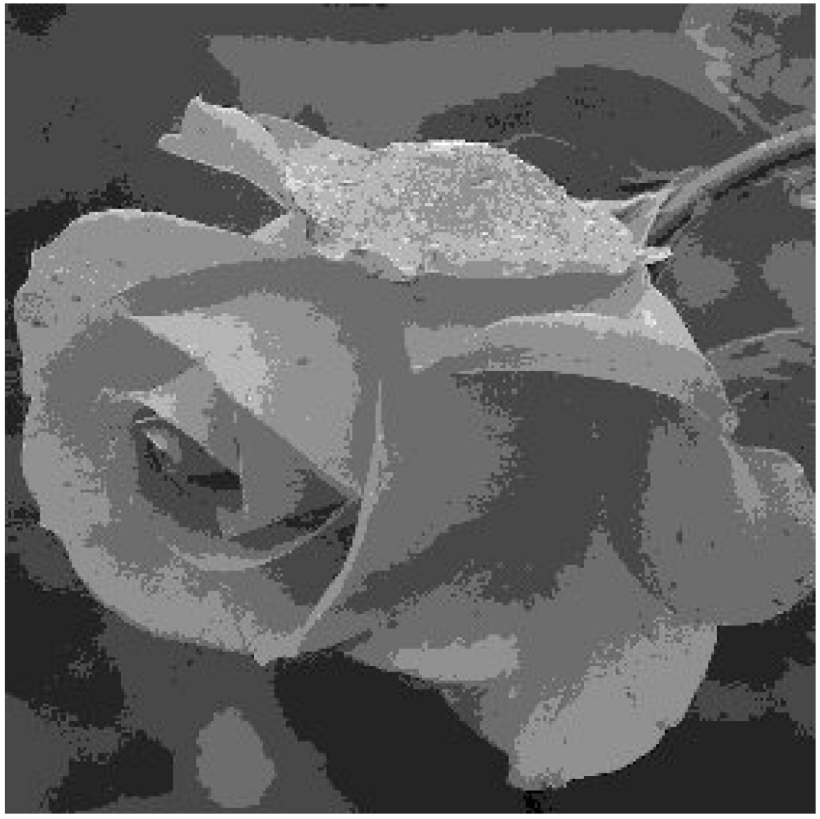
Quantizing Images

8 bit image

4 bit image

Quantizing Images

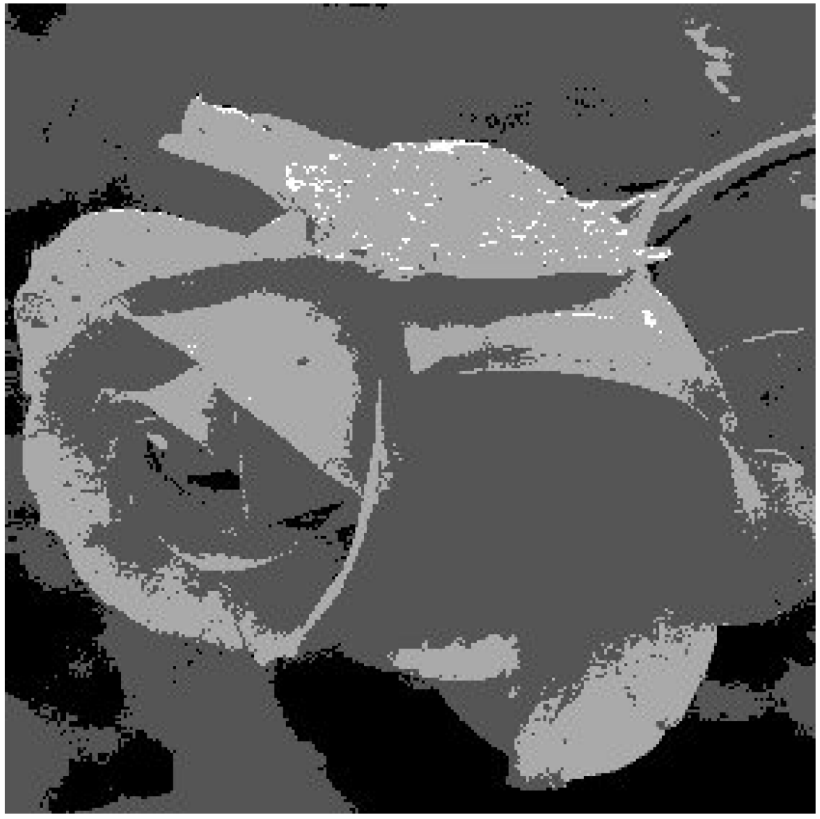
8 bit image



3 bit image

Quantizing Images

8 bit image



2 bit image

Quantizing Images

8 bit image

1 bit image

Check Yourself

What is the most objectionable artifact of coarse quantization?

8 bit image

4 bit image

Dithering

One very annoying artifact is **banding** caused by clustering of pixels that quantize to the same level.

Banding can be reduced by dithering.

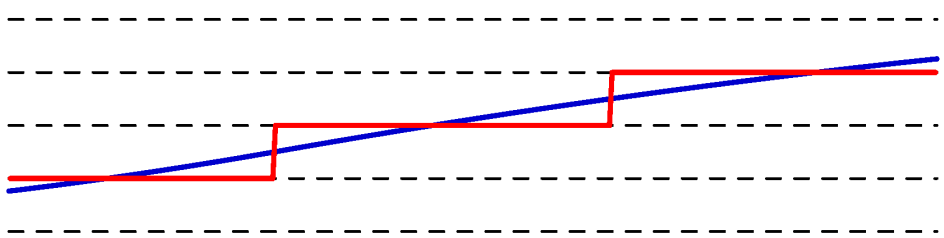
Dithering: adding a small amount ($\pm\frac{1}{2}$ quantum) of random noise to the image before quantizing.

Since the noise is different for each pixel in the band, the noise causes some of the pixels to quantize to a higher value and some to a lower. But the average value of the brightness is preserved.

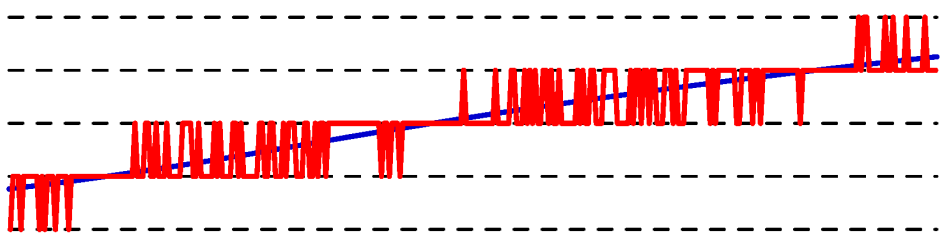
Quantization with Dither

Adding noise (dither) can reduce the perceptual effect of quantization.

Quantization: $y = Q(x)$



Quantization with dither: $y = Q(x + n)$



Quantizing Images with Dither

7 bit image

7 bits with dither

Quantizing Images with Dither

6 bit image

6 bits with dither

Quantizing Images with Dither

5 bit image

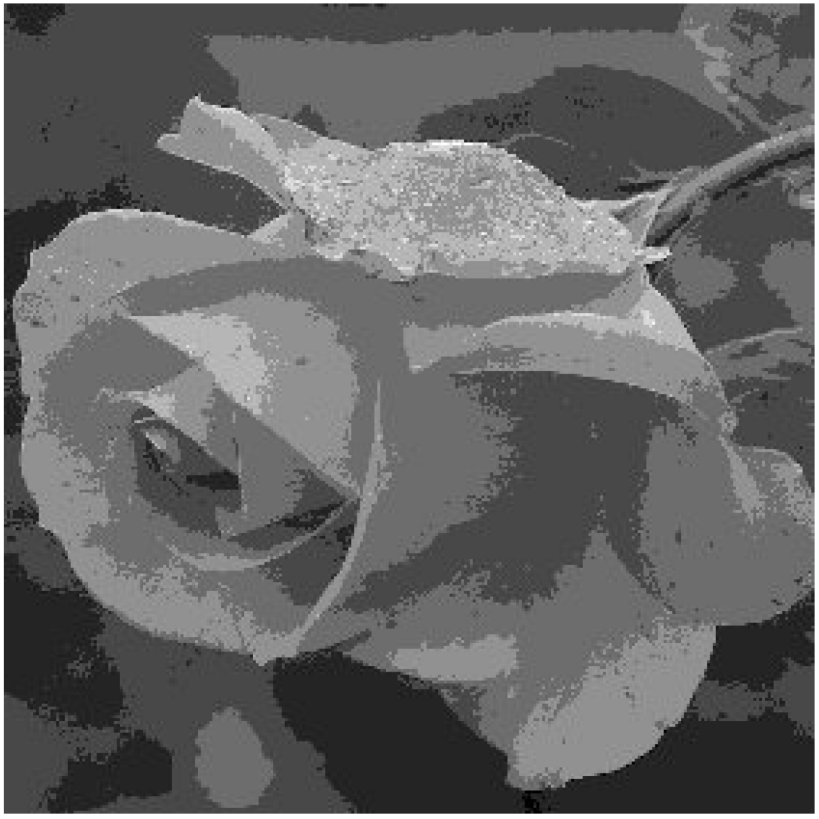
5 bits with dither

Quantizing Images with Dither

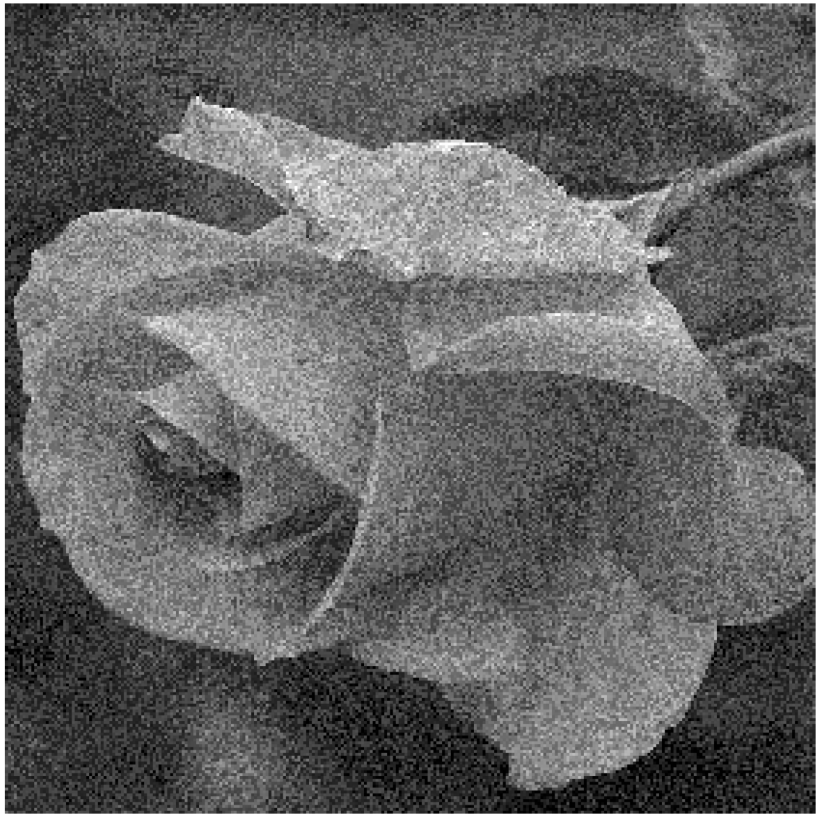
4 bit image

4 bits with dither

Quantizing Images with Dither

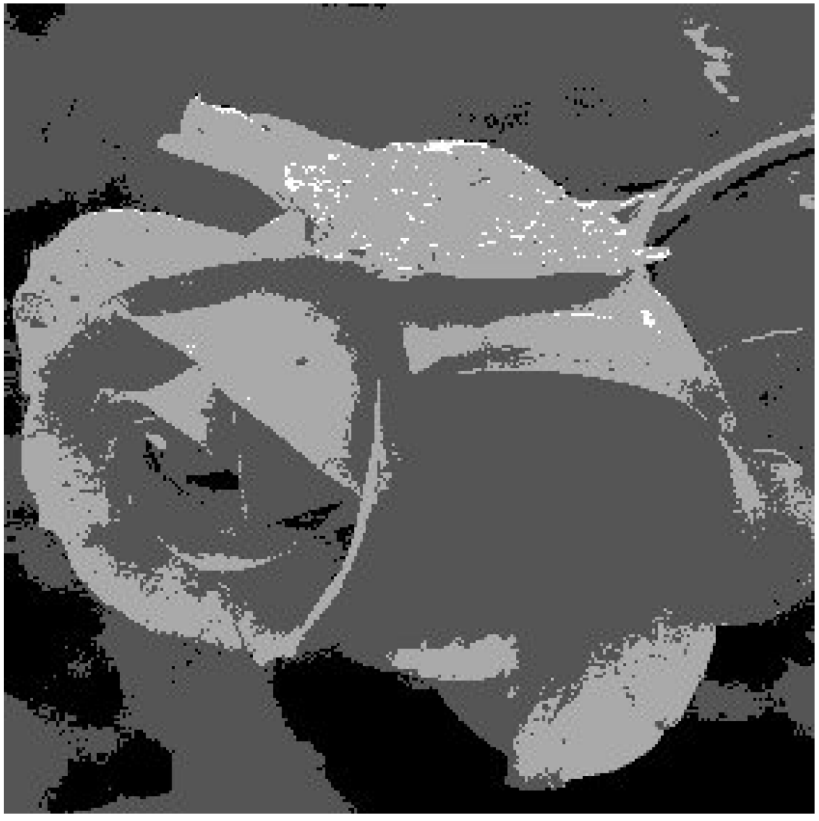


3 bit image

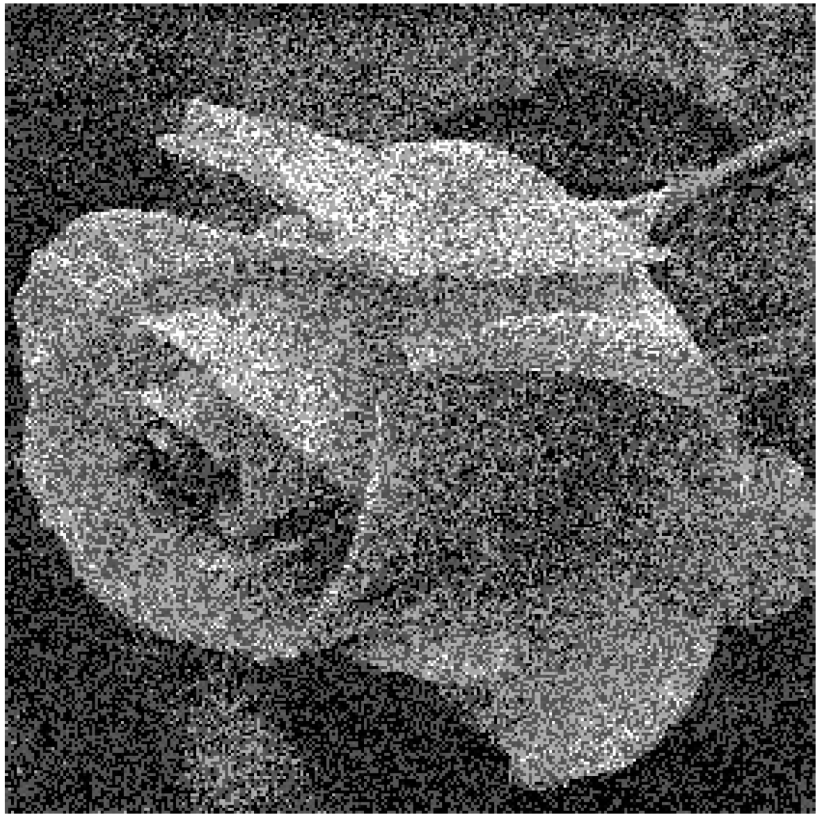


3 bits with dither

Quantizing Images with Dither



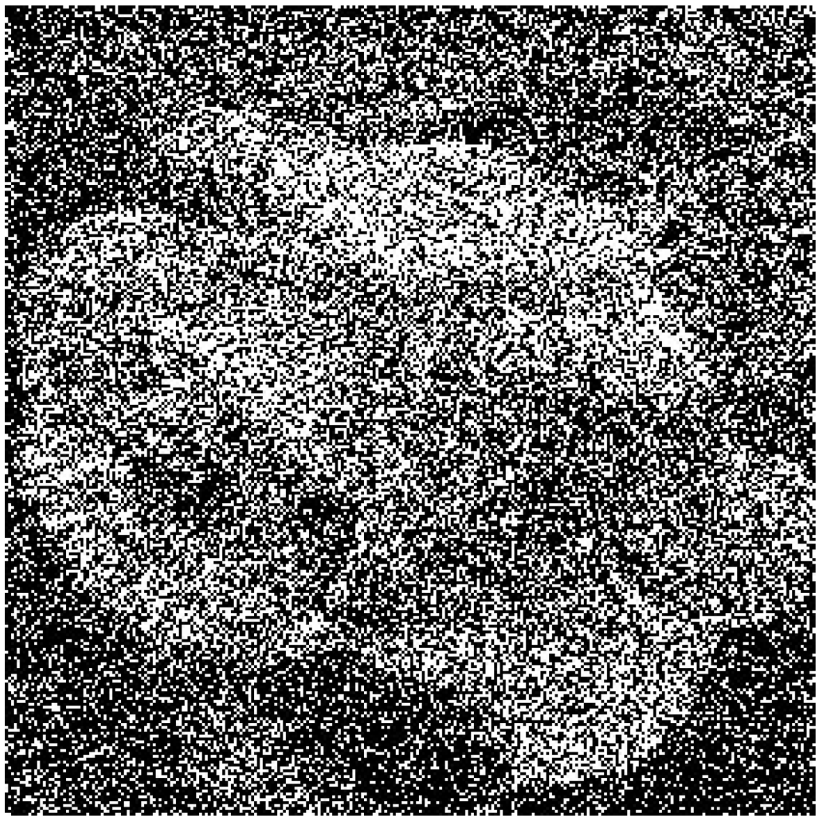
2 bit image



2 bits with dither

Quantizing Images with Dither

1 bit image

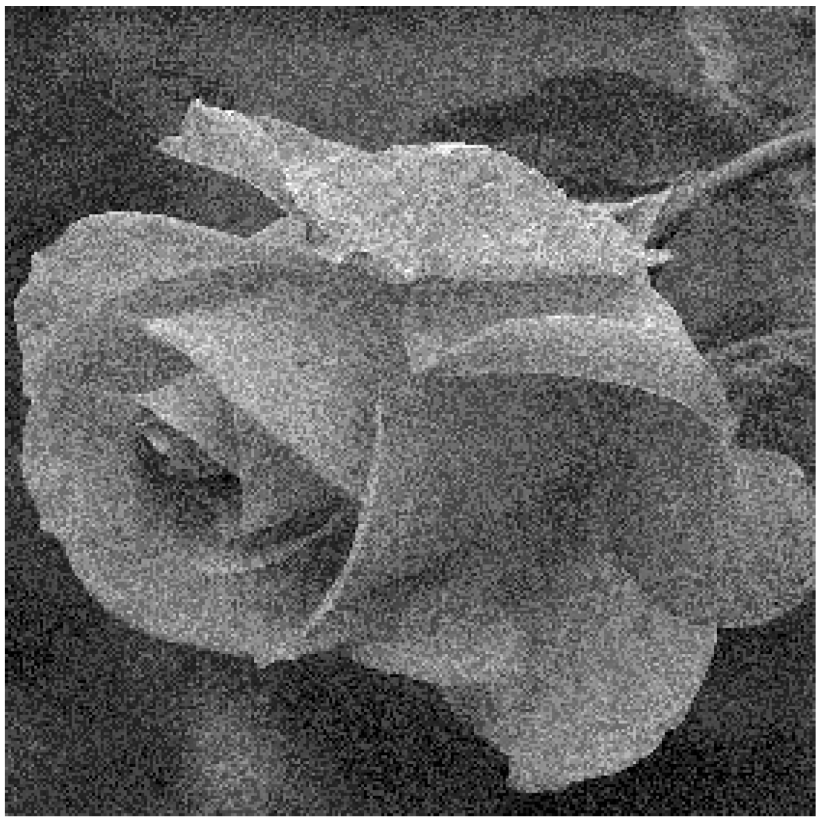


1 bit with dither

Check Yourself

What is the most objectionable artifact of dithering?

3 bit image



3 bit dithered image

Robert's Technique

One annoying feature of dithering is that it **adds noise**.

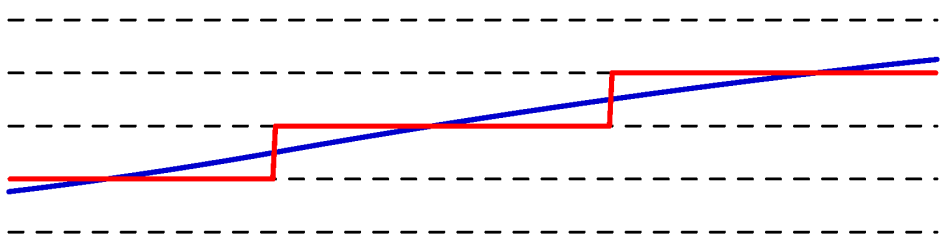
The noise can be reduced using Robert's technique.

Robert's technique: add a small amount ($\pm\frac{1}{2}$ quantum) of random noise before quantizing, then subtract that same amount of random noise.

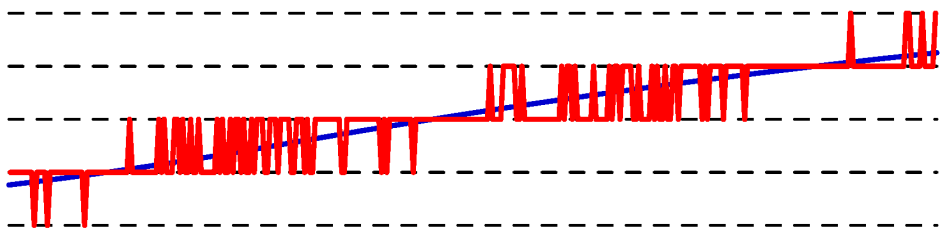
Robert's Technique

Robert's technique decorrelates quantization errors from the signal.

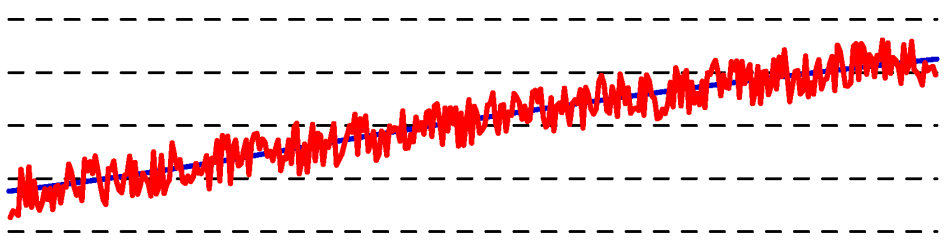
Quantization: $y = Q(x)$



Quantization with dither: $y = Q(x + n)$



Quantization with Robert's technique: $y = Q(x + n) - n$



Quantizing Images with Robert's Method

7 bits with dither

7 bits with Robert's method

Quantizing Images with Robert's Method

6 bits with dither

6 bits with Robert's method

Quantizing Images with Robert's Method

5 bits with dither

5 bits with Robert's method

Quantizing Images with Robert's Method

4 bits with dither

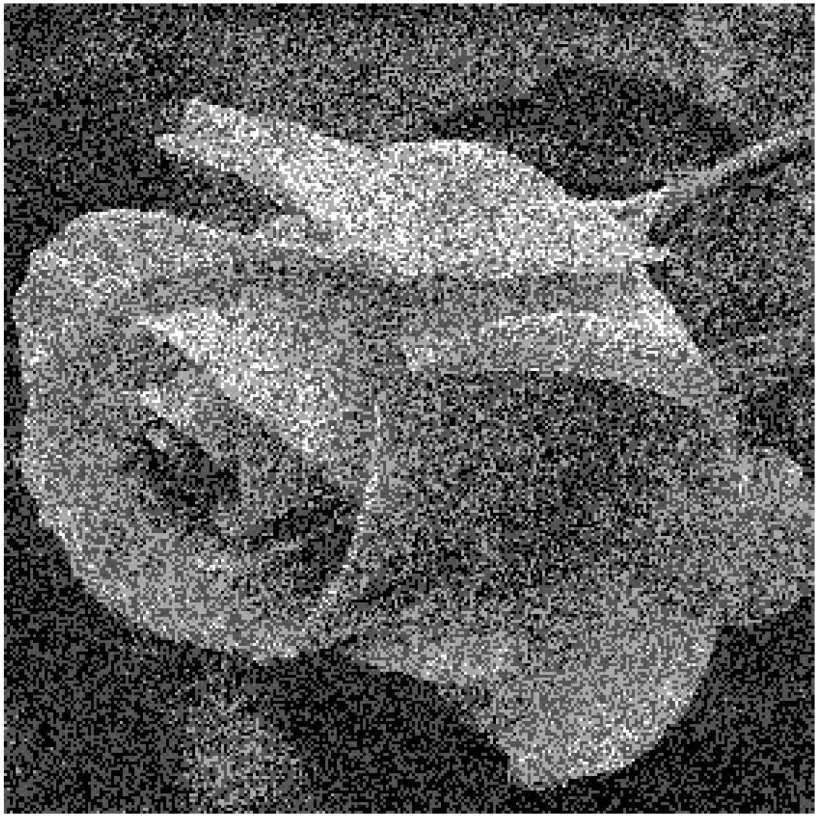
4 bits with Robert's method

Quantizing Images with Robert's Method

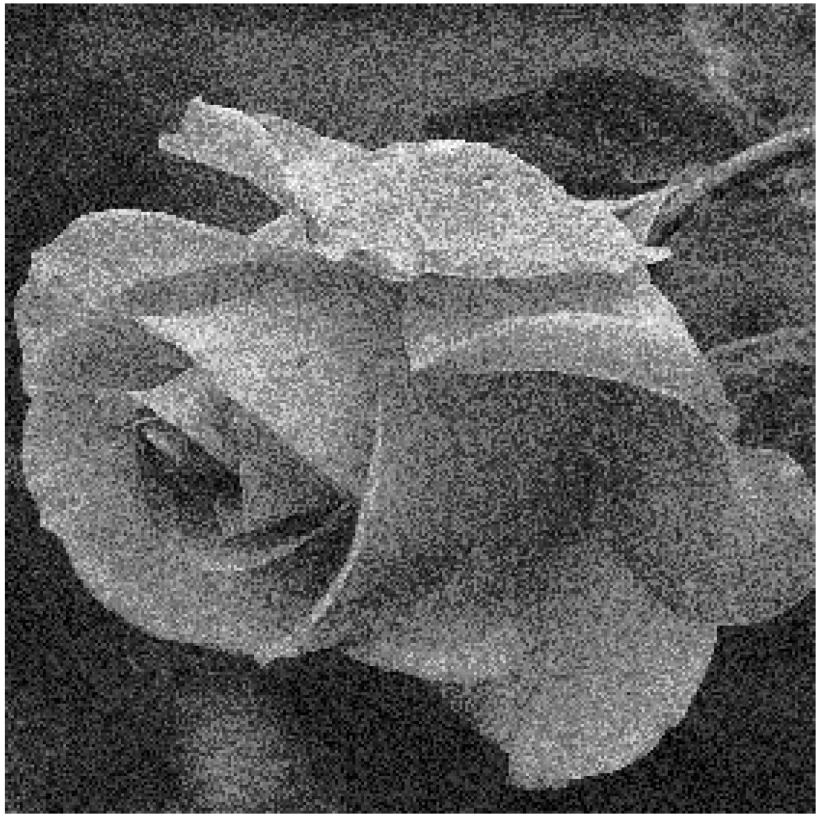
3 bits with dither

3 bits with Robert's method

Quantizing Images with Robert's Method

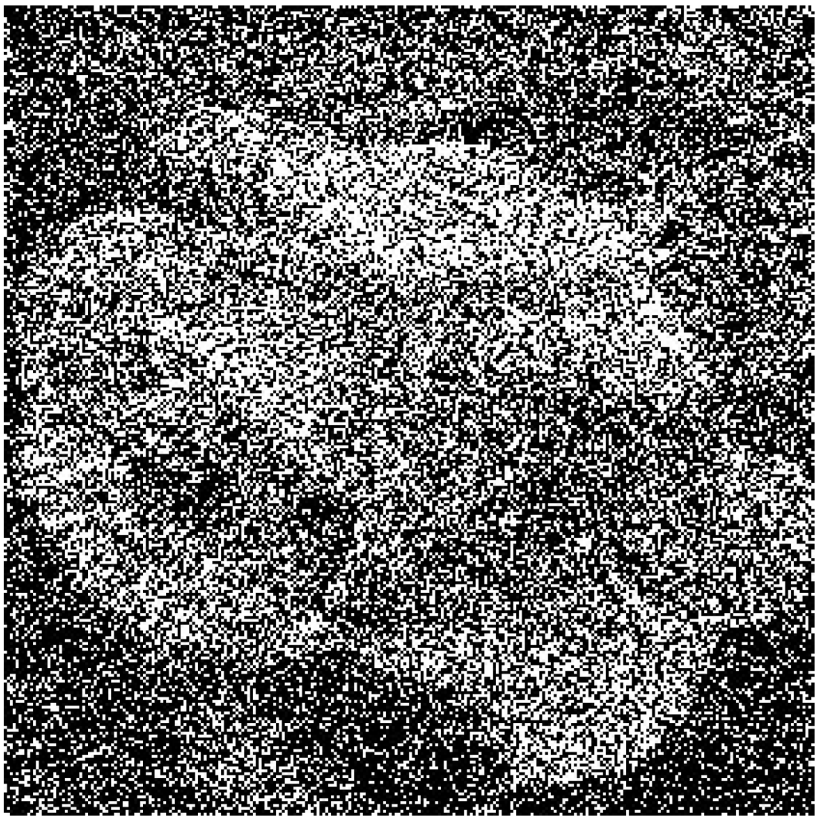


2 bits with dither

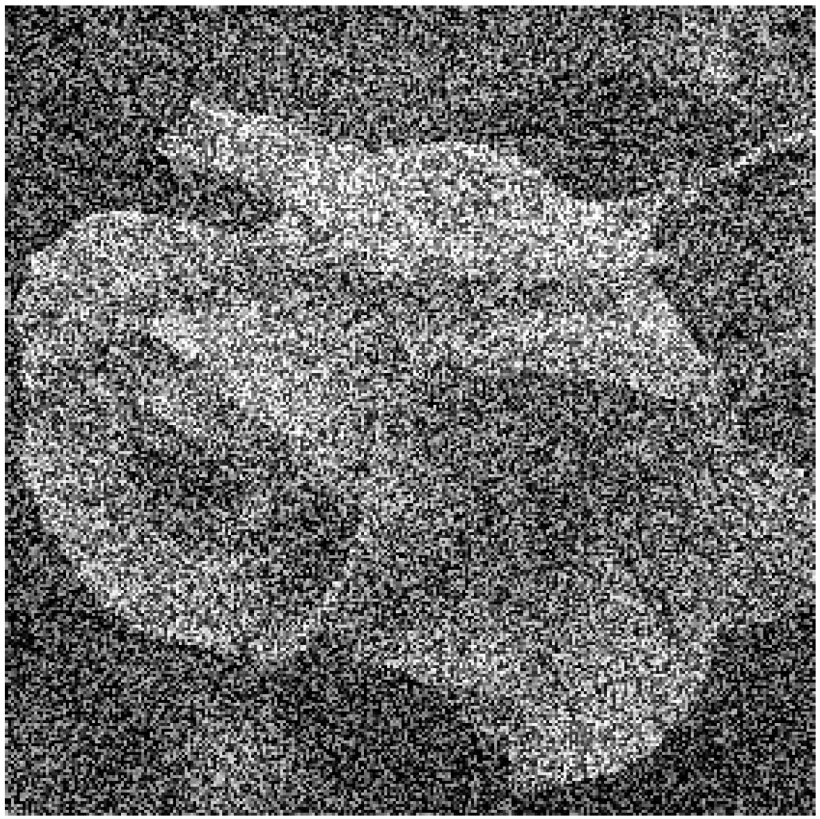


2 bits with Robert's method

Quantizing Images with Robert's Method



1 bits with dither



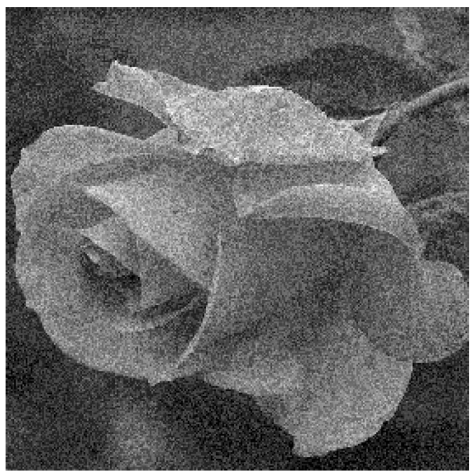
1 bit with Robert's method

Quantizing Images: 3 bits

8 bits

3 bits

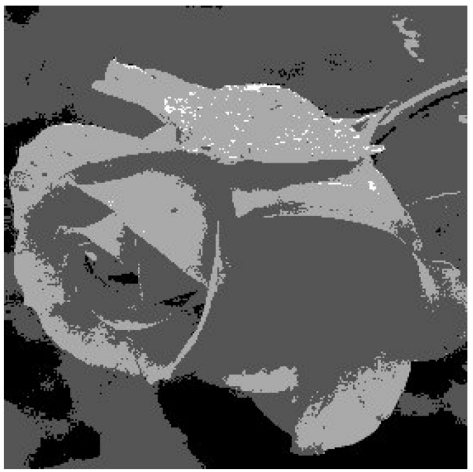
dither



Robert's

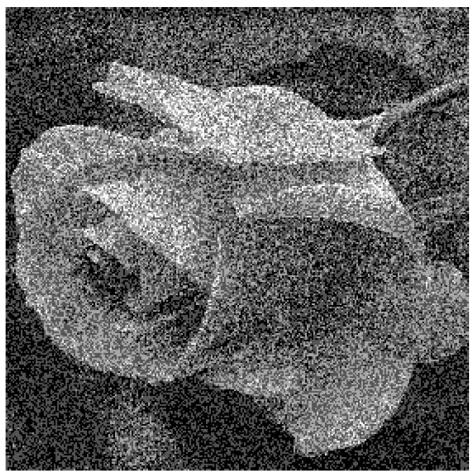
Quantizing Images: 3 bits

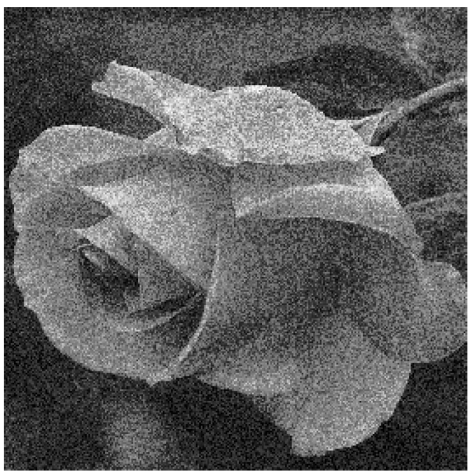
8 bits



2 bits

dither





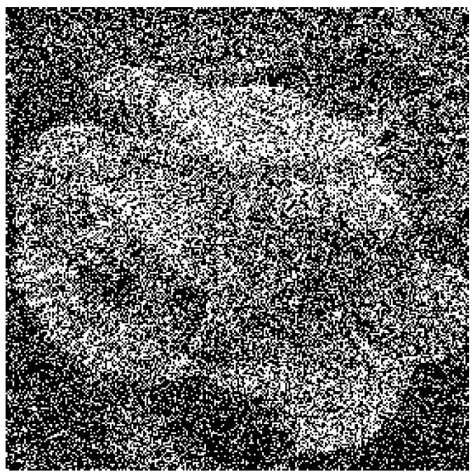
Robert's

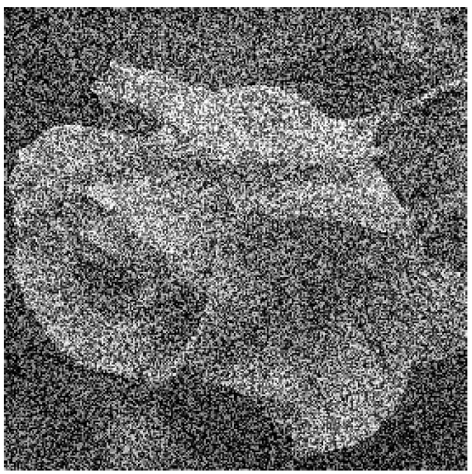
Quantizing Images: 3 bits

8 bits

1 bit

dither

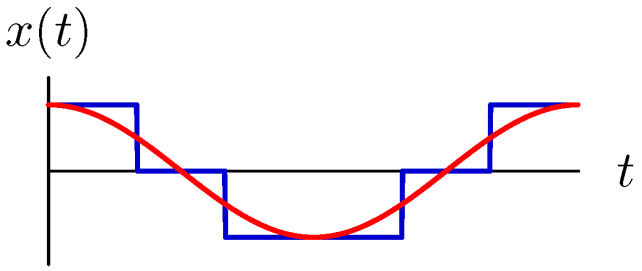


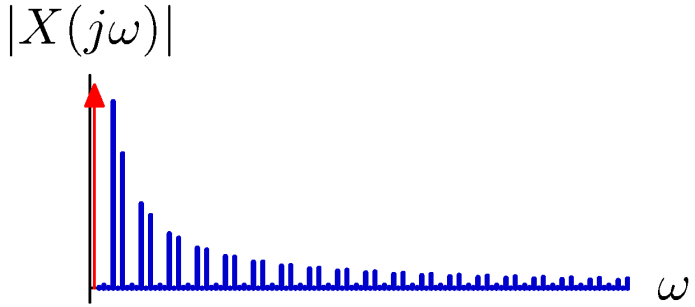


Robert's

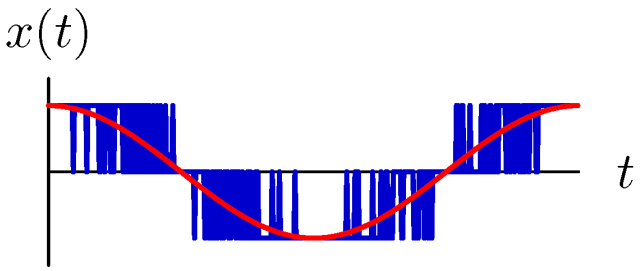
Fourier Analysis

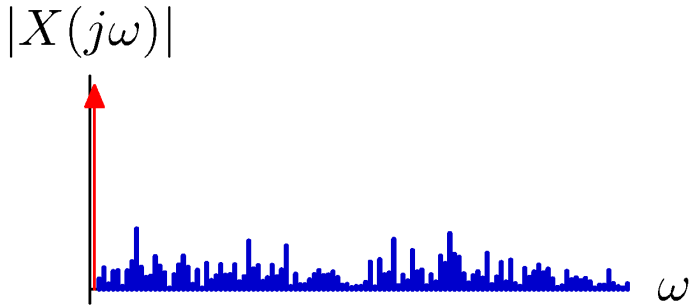
Quantization



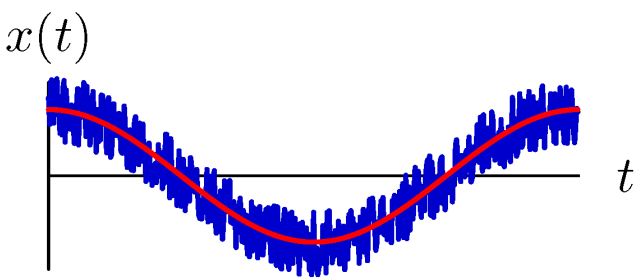


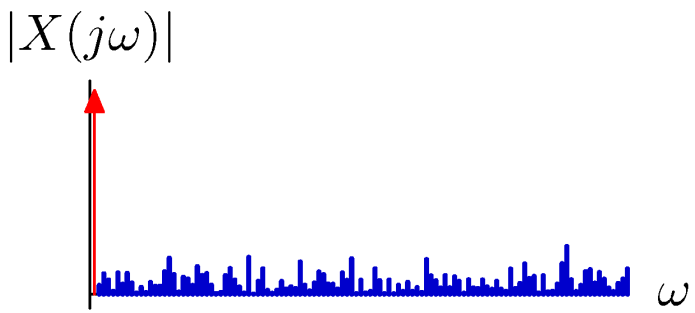
Dither





Robert's Technique





Assignments

- Reading Assignment: Ch. 7